A Semantic Validity in Visual Grounding

The distinction between accurate localization and valid reference motivates the paired evaluation developed below. Appendix A clarifies this distinction and its empirical basis; Appendix B specifies benchmark construction and scoring; Appendix C provides the complete unfiltered results; and Appendix D details the mitigation settings and ablations.

A.1 Localization and Referential Validity

Accurate localization alone does not establish that a model understood the referring expression. A box establishes geometry, whereas a valid reference requires an instance satisfying the complete expression. A model can locate a visible cup accurately and still assign that cup to an unsupported attribute or relation. With the notation of Section 3.1, geometric overlap with b^* does not imply $z(\hat{b}) \in \mathcal{R}(I, q)$. For an unsupported expression, $\mathcal{R}(I, q) = \varnothing$, so any returned region is an invalid assignment. This motivates reporting positive localization and negative-query emission separately. Conditional box reuse (CBR) examines whether the invalid assignment persists in image groups whose paired positive was correctly grounded. It is a behavioral diagnostic, not a claim about the model's internal reasoning.

A.2 Relation to Existing Evaluation

Existing hallucination and grounding benchmarks address complementary aspects of semantic validity. RefCOCOg supplies supported referring expressions and target regions (Mao et al., 2016). CHAIR evaluates hallucinated object mentions in captions, POPE probes object existence, and AMBER covers several hallucination dimensions (Rohrbach et al., 2018; Li et al., 2023; Wang et al., 2023). FineCops-Ref already studies compositional negatives and rejection; CSR and KnowDR-REC further examine counterfactual segmentation and knowledge-conditioned grounding (Liu et al., 2024; Li et al., 2026; Jin et al., 2026). KnowDR-REC also reports a positive-correct-conditioned negative–positive overlap audit, which is a direct antecedent of this style of conditional diagnosis. Thus, unsupported references, rejection, and conditional overlap are established concerns.

Our diagnostic contribution is to hold the image and supported anchor fixed across four edited negatives and compare independently submitted verification and grounding requests, with query-free captions as a separate view. EOH concerns a required but absent object; ROH concerns visible objects that fail the complete attribute or relation constraint. CBR conditions this comparison on successful positive grounding. Complete per-model results are reported once in Appendix C; their role is to test this distinction rather than redefine earlier benchmarks.

A.3 Empirical Basis for the Diagnosis

Three observations motivate examining invalid assignments separately from failures of positive localization or differences between unrelated examples. First, General MLLMs and Grounding-Specific MLLMs have similar positive mIoU (0.4331/0.4240) but different direct-grounding FGR (37.64/68.83%). Second, the query-matched analysis reports explicit verification rejection alongside direct-grounding emission on 24.78% of matched negative records across 13 models (Appendix C.9). Third, relation CBR exceeds object CBR in every evaluated model, even after conditioning on correct positive grounding (Appendix C.7).

These findings distinguish successful localization from valid assignment. They do not identify a causal effect of RL: the model families differ in backbones, supervision, and interfaces. Nor does matching independently generated responses establish a contradiction within a single response. Appendix B.5 separates these events, and Appendix C.9 explains why aggregate rate gaps alone cannot count query-level mismatches.

B S²M-Bench: Construction and Evaluation Protocols

We construct unsupported expressions from shared positive anchors and validate them against the full image. The evaluation then distinguishes task-specific output states and fixes the metric denominators, so that annotation validity and model behavior can be assessed separately.

B.1 Shared Anchors and Semantic Edits

To control the image and target when comparing existential and referential errors, each of the 1,000 image groups retains a RefCOCOg image, one supported expression, and its target box (Mao et al., 2016). Four edited negatives share this anchor, giving 4,000 positive–negative pairs and

5,000 distinct image–expression items. The positive item is shared rather than counted as four distinct positives. Existing negatives are reused only when their positive expression matches the anchor; an MLLM supplies missing types.

Object edits replace an object mention with an absent category; association edits introduce a plausible but absent required object. Both are EOH, including when the missing object is the reference object in a relation. Attribute and relation edits preserve the required objects' visibility but invalidate a property, predicate, or participant binding, and instantiate ROH. Association is a construction category, not a claim that co-occurrence proves visual presence; POPE provides related motivation for examining object-existence errors (Li et al., 2023).

B.2 Image-Wide Validation and Human Review

Validation covers the full image to exclude alternative targets that might satisfy an edited expression. EOH review checks that the required object is absent throughout the image. ROH review checks that the required objects are visible and that no alternative target satisfies the edited expression. Relation checks include reference identity, same-class alternatives, exchanged participant roles, and depth or direction ambiguity. The target box annotates only the supported positive.

The `qwen-max` annotator supplies semantic units and risk flags after the four edits; GPT and Gemini cross-check the generated annotations. Four researchers experienced in MLLM dataset construction review all 5,000 items. Each item has one primary reviewer; image assignments overlap, so multiple reviewers can inspect different items from one image. Review covers image–text correspondence, positive validity, negative invalidity, EOH/ROH type, and the target region. Reviewers resolve flagged disagreements before accepting annotations. This procedure establishes complete item coverage and overlapping image assignments; it does not provide a count of independently double-labeled items or a pre-adjudication agreement coefficient. Structural checks enforce one shared positive and all four aligned negative types per group.

B.3 Output Parsing and Coordinate Conventions

Scoring distinguishes malformed outputs and refusals from valid geometric predictions. Coordinate conversion is model-specific and precedes clipping and overlap. UniVG-R1 and Visual-RFT emit normalized coordinates in $[0, 1000]$; these are scaled by the actual image width and height. Parsing requires a recognized output structure. Emission, valid geometry, semantic decisions, format failures, and missing executions are distinct fields (Table A1). FGR counts parsed box emissions over all negative queries, including degenerate boxes as emissions with zero IoU. Empty, rejected, or invalid positive predictions contribute zero to all-positive mIoU. A refusal accompanied by a parsed box still counts as an emission. Missing executions require recovery for completeness and must not be interpreted as successful refusals.

Output state	Scoring treatment
Parsed, nondegenerate box	Emission $e = 1$; score geometric IoU.
Parsed box with invalid/degenerate geometry	Emission $e = 1$; IoU and reuse are zero.
Explicit refusal with no parsed box	Emission $e = 0$; zero positive IoU and reuse.
Refusal text plus a parsed box	Record both signals; $e = 1$ and check their contradiction.
Malformed text with no parsed box	Record format failure separately from refusal; retain the query in grounding denominators.
Missing execution or response	Record missing execution or response separately; completeness requires recovered outputs.

Table A1: Grounding-output scoring rules.

B.4 Metric Denominators and Aggregation

Fixed denominators preserve the evaluated population when predictions are filtered. Each model has 1,000 evaluated positive queries and 4,000 negatives: 1,000 per edit type and 2,000 per EOH/ROH stratum. CBR uses the originally correct positive set at IoU ≥ 0.5 and tests negative-to-positive box overlap at IoU ≥ 0.8 (Equation 3). The four types share the same denominator within a model and task. Every paired negative, including a refusal, stays in that denominator. Empty or invalid boxes have zero IoU; CBR is undefined if no positive qualifies.

Direct and joint grounding each use their own

unfiltered positive-correct set. Within a mitigation comparison, both eligibility and positive reference predictions are frozen before filtering. Equal-model means average each model's ratio, rather than pooling eligible groups across models. Correct-grounding retention divides retained originally correct positives by all originally correct positives; it is not the fraction of all positive queries receiving a box. All-positive mIoU and full-denominator FGR accompany retention to expose the cost of rejection.

Verification FAR uses valid negative support judgments, excluding abstentions and parse failures; balanced accuracy averages positive acceptance and negative rejection within valid judgments. These denominators differ from grounding's full negative-query denominator.

B.5 Caption Assertions and Cross-Task Alignment

The caption generator receives only the image, without the challenged expression. For semantic evaluation, each model's 1,000 descriptions are checked against four annotated negative units per image, yielding 2,000 EOH and 2,000 ROH judgments. The full set of caption-assertion judgments is evaluated with `qwen-max` and cross-checked by GPT and Gemini. Four researchers then cross-review all judgments.

Object mentions are scored with entity binding, attributes with their target entity, and relations with their predicate and participant roles. Affirmative paraphrases count; negations, quotations, and mere lexical co-occurrence do not. Semantic assertion rates differ from the object-level hallucination and target-coverage measures retained in Table A10. CHAIR and AMBER motivate object/claim-level evaluation (Rohrbach et al., 2018; Wang et al., 2023); the study's annotated-unit entailment protocol is a separate measure.

Cross-task records. Records are matched by model and query identifier to compare verification judgments with grounding outputs for the same expression. Explicit support, unsupported judgments, and abstentions remain separate when compared with region emission. A verification rejection paired with direct-grounding emission compares independent task responses. Within joint grounding, an explicit unsupported judgment accompanied by a box is instead a contradiction within one response. These are distinct events; neither should be estimated from an aggregate rate difference.

B.6 Statistical Scope and Sensitivity

The scope of uncertainty depends on the resampling unit. The 13-model mitigation tables report point estimates. For image-group resampling, the specified sampling unit is the source image ID, grouping all model outputs, the positive and four negatives, fold membership, and policy decisions. Numerators and denominators are recomputed before averaging paired changes. With trained strategies fixed, this targets conditional image-sampling uncertainty, not retraining variability.

The matched-retention analysis reports both source-image and model-bootstrap intervals from 10,000 resamples (Appendix D.6). Image resampling keeps the 12 models paired; model resampling characterizes variation across those models.

C Extended Evaluation and Analysis

This section reports the complete unfiltered evaluation and examines semantic difficulty, conditional region reuse, and changes induced by the task interface. We first describe the evaluated models and the scope of their outputs, then relate these factors to the observed semantic–spatial mismatch. Mitigation settings and results appear in Appendix D.

Repeated inference and aggregation. All evaluation metrics reported in the main paper and appendix, including mitigation and ablation results, are arithmetic means over three inference runs on the same evaluation image groups. Metrics are computed for each run and then averaged. Cross-model summaries give equal weight to each model. Hardware and execution details appear in Appendix D.13.

C.1 Evaluated Models and Comparison Scope

The evaluated set comprises four General MLLMs and nine Grounding-Specific MLLMs. The descriptions below link each model family to its original publication. The models differ in backbones, task mixtures, supervision, and output interfaces. InternVL3.5 itself incorporates RL (Wang et al., 2025); the General MLLMs label does not mean RL-free. Family means are descriptive comparisons over this selected set, not controlled estimates of an RL effect. The cited sources identify

the upstream methods rather than isolating any one training factor.

Qwen3-VL-8B (Bai et al., 2025a) is an 8B multimodal language model from the Qwen3-VL family. It combines image understanding with instruction following, visual question answering, and language-based reasoning over visual content.

Qwen2.5-VL-7B (Bai et al., 2025b) is a 7B model from the Qwen2.5-VL family. It supports image understanding, visual question answering, and multimodal instruction following with language responses grounded in the image.

InternVL3.5-8B (Wang et al., 2025) is an 8B model from the InternVL3.5 multimodal language model family. The family targets general visual perception and multimodal reasoning, including image interpretation and language-conditioned understanding.

LLaVA-OV-7B (Li et al., 2025) is a 7B model from the LLaVA-OneVision family. It is intended for general image understanding and multimodal instruction following, with language outputs conditioned on visual and textual inputs.

LENS-3B (Zhu et al., 2026) is a 3B multimodal model with an RL-adapted visual grounding configuration. The model belongs to the class of compact multimodal systems adapted to reason about referring expressions and select corresponding regions.

Orsta-7B (Ma et al., 2025) is a 7B multimodal model with an RL-adapted visual grounding configuration. It is designed for language-guided visual reasoning in which an expression is associated with one or more candidate image regions.

Seg-R1-7B (You and Wu, 2025) is a model for reasoning-guided visual segmentation and grounding. It combines multimodal reasoning with region-level prediction, supporting visual queries that require identifying and grounding the described content.

Seg-Zero-7B (Liu et al., 2025a) is a reasoning-chain-guided segmentation model trained with cognitive reinforcement. Its reasoning module is trained to guide segmentation decisions, connecting intermediate visual reasoning with the selection of image regions.

TreeVGR-7B (Wang et al., 2026) is a 7B multimodal model with an RL-adapted visual grounding configuration. It represents referring expressions with a reasoning-oriented grounding process that maps language descriptions to visual regions.

UniVG-R1-7B (Bai et al., 2025c) is a reasoning-guided universal visual grounding model trained with reinforcement learning. The framework uses supervised reasoning traces and reinforcement learning to support grounding across different visual referring tasks.

Vision-R1-7B (Zhan et al., 2025) is a multimodal model with an RL-adapted visual reasoning and grounding configuration. It is intended to combine explicit visual reasoning with language-conditioned region prediction for image referring queries.

VisionReasoner-7B (Liu et al., 2026) is developed for visual reasoning and grounding. The model emphasizes reasoning over visual scenes before producing language responses or spatial assignments for the requested content.

Visual-RFT-7B (Liu et al., 2025b) is a visual reinforcement fine-tuning approach for visual perception and grounding. It uses verifiable rewards, including grounding quality, to improve the connection between perception and predicted regions.

Model	System handling
Qwen2.5-VL, Qwen3-VL, LLaVA-OV, TreeVGR, Orsta (OneRL), Vision-R1	S1
LENS, UniVG-R1	S2
Visual-RFT, Seg-R1, Seg-Zero, VisionReasoner	No explicit system message
InternVL3.5	Separate chat adapter[a]

Table A2: System-message handling in the evaluator. S1 and S2 are reproduced below. [a]The adapter passes the image and user instruction to the model's `chat` method; it does not pass the configured S1 string. The model-internal system text is not established by this adapter.

C.2 Prompts and Response Formats

Message structure. We separate the system message from the task instruction in the user message. The evaluator places the image and task instruction together in the user message; the expression is supplied for verification, direct grounding, and joint grounding, but omitted for query-free captioning. Below we reproduce the Qwen2.5-VL templates, with `{expr}` denoting the substituted expression. Table A2 records the model-specific system-message handling. The accompanying source supplement retains the full model-specific task templates.

Table A3: **System S1 Prompt.**

System message
You are a careful visual grounding assistant. Use only visible image evidence. Follow the requested output format exactly.

Table A4: **System S2 Prompt.**

System message
You are a helpful assistant.

Table A5: **Verification Prompt.**

User message
Task: Decide whether the referring expression strictly matches a visible target in the image. Referring expression: `"{expr}"` The object identity, number, attributes, colors, and spatial relations must all match. Answer with exactly one word: yes or no.

Table A6: **Direct Grounding Prompt.**

User message
Task: Decide whether the referring expression strictly matches a visible target in the image, then localize it if it exists. Referring expression: `"{expr}"` If the target exists, return exactly one bounding box in absolute image pixels as [x1, y1, x2, y2]. If the target does not exist, return exactly: not found. Do not output explanations.

Table A7: **Query-free Captioning Prompt.**

User message
Describe this image in two or three sentences. Mention only what is clearly visible. Do not guess or infer. Output the description only.

Table A8: **Joint Grounding Prompt.**

User message

Task: First describe the image in one concise sentence. Then check whether `"{expr}"` strictly matches a visible target.
Step 1 — Describe the image concisely.
Step 2 — Is there `"{expr}"` in the image?
The object identity, number, attributes, colors, and spatial relations must all match.
If yes, provide the bounding box in absolute image pixels as [x1, y1, x2, y2].
If no, output "not found".
Return JSON only:
`{"description":"one concise sentence", "exists":"yes", "bbox":[x1,y1,x2,y2]}`
`{"description":"one concise sentence", "exists":"no", "bbox":"not found"}`
Do not output explanations.

Model-specific formats. The full templates are not identical across models. The LENS, Seg-Zero, VisionReasoner, TreeVGR, and UniVG-R1 interfaces request task-specific `<think>` and/or `<answer>` wrappers; Visual-RFT uses a `Question:` form. For captioning, the standard template above is used for the standard family and as an explicit UniVG-R1 override. The other reasoning-style caption templates request a final description in `<answer>` tags. These are output-format instructions in the user message, not additional system messages. This protocol variation is another reason to interpret model-family comparisons descriptively.



(a) COCO 525342



(b) COCO 560009

Figure A1: Images for the saved responses in Appendix C.3. Verification, direct grounding, and captioning use (a); the joint-grounding example uses (b).

C.3 Representative Evaluation Examples

The following Qwen2.5-VL examples are verbatim saved responses, with line wrapping adjusted for display. They include a correct rejection as well as an unsupported spatial assignment; they are qualitative examples rather than an estimate of error frequency. The first three use COCO image 525342. The joint-grounding example uses image 560009 and therefore is not a fourth response to the same image.

Verification example. Input expression: the green orange in front of the apples.
Saved response: `no`.

Direct-grounding example. Input expression: the green orange in front of the apples.
Saved response: `not found.` The final period belongs to the raw response and is accepted by the refusal parser.

Query-free caption example. Input: the image and the captioning instruction above, with no referring expression.
Saved response:

Saved response
The image shows a close-up of several fruits, including two oranges and four apples, arranged on a white surface with green leaves partially visible beneath them. The lighting highlights the textures of the fruits' skin.

Joint-grounding example. Input expression: the car in front of the fence.

Saved response:

Saved response

```
```json
{"description":"Two men in suits are
conversing near a decorative fence with
people in the background.",
"exists":"yes",
"bbox_2d":[178,134,386,450]}
```
```

The saved annotation marks the expression unsupported. Although the description mentions men and a fence, the response asserts existence and emits a region for the car query. The `bbox_2d` key and Markdown fence are preserved from the response; they differ from the requested `bbox` format. ASV acts on the spatial candidate and does not rewrite this upstream description or existence string.

C.4 Model-Family Profiles

Measure	General MLLMs	Grounding-Specific MLLMs
Models	4	9
False acceptance (%)	29.44	36.74
FGR (%)	37.64	68.83
Positive success (%)	45.0	42.9
mIoU	0.4331	0.4240
Caption error (%)	3.80	4.51
Target coverage (%)	50.70	57.89
Joint FGR (%)	47.34	48.66
Joint success (%)	33.75	36.42
Joint mIoU	0.3551	0.3679

Table A9: Equal-model family means. Caption metrics evaluate query-free object claims.

General MLLMs and Grounding-Specific MLLMs have similar positive grounding success (45.0/42.9%) and mIoU (0.4331/0.4240), while direct FGR differs by 31.19 percentage points (37.64/68.83%). Figure A2 shows the per-model relationship without collapsing the two cohorts' denominators.

C.5 Caption Scope and Explicit Rejection

The scope of a caption differs from that of an explicit grounding request. Query-free captions select their own content; verification and grounding must respond to a supplied expression. In Table A10, object hallucination ranges from 2.00–5.20%, while target coverage ranges from 29.00–62.60%. A caption can omit a difficult target or constraint. Its low object error therefore does not establish that the model will reject an independently submitted false request. Semantic caption

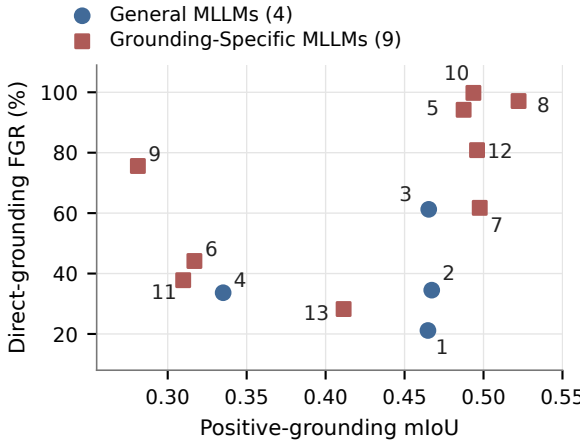


Figure A2: Unfiltered localization and false grounding. Each point is a model, numbered in the model order of Appendix C.1. mIoU uses 1,000 positives and FGR uses 4,000 negatives per model.

assertion rates, reported by EOH/ROH in Table 2, additionally test the annotated negative units under Appendix B.5.

MLLM	BAcc	FAR	Obj. err.	Coverage
Qwen3-VL-8B	82.85	23.70	4.40	57.00
Qwen2.5-VL-7B	80.93	22.35	4.40	61.40
InternVL3.5-8B	79.48	26.25	4.40	55.40
LLaVA-OV-7B	75.78	45.45	2.00	29.00
LENS-3B	70.82	37.32	4.20	53.60
Orsta-7B	80.78	26.45	4.80	60.20
Seg-R1-7B	78.60	20.80	5.00	61.60
Seg-Zero-7B	76.32	45.14	5.00	57.00
TreeVGR-7B	66.78	66.23	3.80	49.00
UniVG-R1-7B	73.08	35.62	4.20	60.00
Vision-R1-7B	81.13	30.35	5.20	61.60
VisionReasoner-7B	76.87	42.24	4.60	62.60
Visual-RFT-7B	80.63	26.55	3.80	55.40

Table A10: Expression verification and query-free captioning (%). BAcc/FAR use valid support judgments; object error and target coverage score query-free descriptions. These caption metrics differ from semantic-unit assertion rates.

C.6 Semantic Difficulty across Models

The per-model breakdown in Table A11 shows that the EOH/ROH difference persists within individual models. Verification FAR and direct-grounding FGR are higher for ROH than EOH in all 13 models. The equal-model direct FGR is 48.11/70.36% for EOH/ROH. This supports the paper's within-model claim under explicit verification and direct-grounding queries; it is not a claim about every output interface. Figure A3 summarizes the verification contrast by model family: EOH/ROH FAR is 16.63/42.25% for Gen-

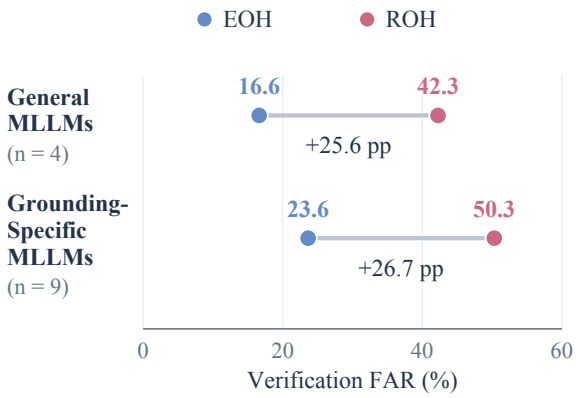


Figure A3: **EOH/ROH difficulty in verification.** Equal-model verification FAR for General MLLMs ($n = 4$) and Grounding-Specific MLLMs ($n = 9$); labels along the connectors show ROH-minus-EOH gaps (percentage points).

eral MLLMs and 23.64/50.31% for Grounding-Specific MLLMs. A small EOH/ROH gap can

MLLM	Verify EOH	Verify ROH	Ground EOH	Ground ROH
Qwen3-VL-8B	11.60	35.80	10.10	32.20
Qwen2.5-VL-7B	11.80	32.90	20.00	49.00
InternVL3.5-8B	14.30	38.20	41.60	80.90
LLaVA-OV-7B	28.80	62.10	18.20	49.10
LENS-3B	25.08	49.60	90.40	98.00
Orsta-7B	15.10	37.80	27.90	60.40
Seg-R1-7B	10.20	31.40	47.00	76.50
Seg-Zero-7B	29.59	60.78	95.30	98.90
TreeVGR-7B	48.95	84.91	62.20	89.00
UniVG-R1-7B	25.35	48.50	99.70	99.90
Vision-R1-7B	16.80	43.90	22.30	53.30
VisionReasoner-7B	28.30	56.18	73.50	88.20
Visual-RFT-7B	13.40	39.70	17.20	39.30

Table A11: False acceptance and direct-grounding FGR (%). Each semantic stratum has 1,000 negative items; verification rates use valid support judgments. Direct-grounding strata cover all 13 evaluated upstream MLLMs.

coexist with almost universal emission: UniVG-R1 has 99.70/99.90% negative emission, alongside 50.8% positive success. The gap alone is therefore insufficient; the absolute rates and positive performance must also be reported.

C.7 Conditional Box Reuse

Conditioning on correct positive grounding allows us to examine false assignments in groups where the model can locate the supported target. Direct-grounding CBR conditions on correctly grounded positives and still shows greater relation than object reuse in all 13 models (Table A13). Each row's four rates share the same eligible positives, including every paired negative refusal. This isolates invalid assignment in groups with successful positive localization, while retaining the conditional scope of the claim. Table A12 summarizes EOH/ROH reuse in both families and grounding tasks. ROH CBR exceeds EOH CBR in every family–task combination, with direct-grounding gaps of 23.02 and 14.79 percentage points for General and Grounding-Specific MLLMs, respectively. Figure A4 visualizes the four-type profile: relation reuse exceeds object reuse within every model, while absolute rates vary substantially across models. The common color scale supports comparison with the mitigation heatmaps in Appendix D. Joint CBR uses each model's joint positive-correct set. Visual-RFT's object CBR is 3.79% in direct grounding and 17.39%

Family	Models	Direct grounding EOH	Direct grounding ROH	Joint grounding EOH	Joint grounding ROH
General MLLMs	4	11.24	34.26	23.45	40.91
Grounding-Specific MLLMs	9	25.41	40.20	14.08	31.20

Table A12: Family CBR (%), extending the paired-grounding analysis of KnowDR-REC (Jin et al., 2026): equal-model means on original positive-correct sets, averaging two edit types per hallucination level.

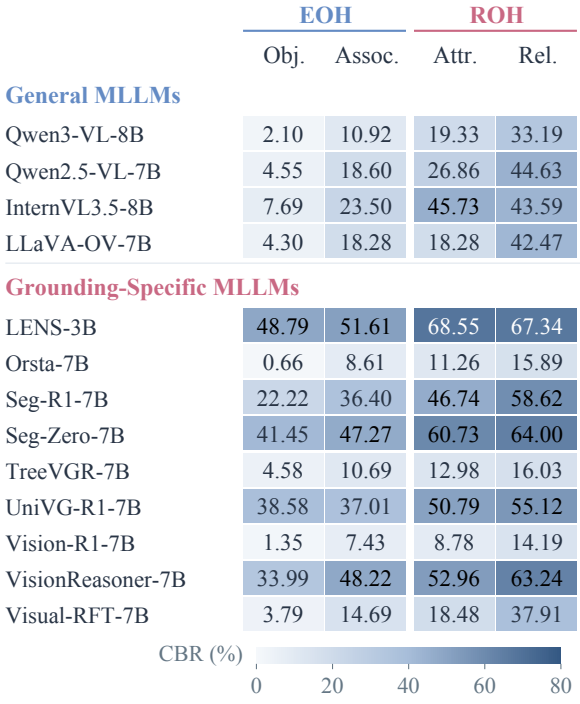


Figure A4: Four-type direct-grounding CBR across all 13 models. Cells show percentages on each model's fixed positive-correct set; lower is better. The four rates in each row share eligibility, while eligibility differs across models (Table A16).

MLLM	Obj.	Assoc.	Attr.	Rel.
Qwen3-VL-8B	2.10	10.92	19.33	33.19
Qwen2.5-VL-7B	4.55	18.60	26.86	44.63
InternVL3.5-8B	7.69	23.50	45.73	43.59
LLaVA-OV-7B	4.30	18.28	18.28	42.47
LENS-3B	48.79	51.61	68.55	67.34
Orsta-7B	0.66	8.61	11.26	15.89
Seg-R1-7B	22.22	36.40	46.74	58.62
Seg-Zero-7B	41.45	47.27	60.73	64.00
TreeVGR-7B	4.58	10.69	12.98	16.03
UniVG-R1-7B	38.58	37.01	50.79	55.12
Vision-R1-7B	1.35	7.43	8.78	14.19
VisionReasoner-7B	33.99	48.22	52.96	63.24
Visual-RFT-7B	3.79	14.69	18.48	37.91

Table A13: Direct-grounding CBR (%). All four types share each model's original positive-correct set, whose eligibility is reported in Table A16. Negative refusals remain in the denominator; source precision is preserved.

MLLM	Obj.	Assoc.	Attr.	Rel.
Qwen3-VL-8B	1.7	7.8	12.1	32.3
Qwen2.5-VL-7B	3.3	17.0	21.7	43.4
InternVL3.5-8B	6.7	15.3	20.0	28.7
LLaVA-OV-7B	66.7	69.1	81.5	87.7
LENS-3B	23.4	18.8	31.2	36.7
Orsta-7B	0.0	10.1	11.5	21.6
Seg-R1-7B	3.9	19.0	23.9	49.3
Seg-Zero-7B	5.6	20.3	23.7	47.4
TreeVGR-7B	0.0	7.1	6.3	13.5
UniVG-R1-7B	25.91	34.41	47.37	55.06
Vision-R1-7B	0.0	8.1	8.1	14.9
VisionReasoner-7B	5.3	29.5	30.9	55.1
Visual-RFT-7B	17.39	24.64	35.27	49.76

Table A14: Joint-grounding CBR (%). All four types share each model's original positive-correct set, whose eligibility is reported in Table A16. Negative refusals remain in the denominator; source precision is preserved.

in joint grounding; relation CBR is 37.91/49.76%. UniVG-R1's relation CBR is 55.12/55.06%, while object CBR changes from 38.58% to 25.91%. Because eligibility can change across tasks, these are task-specific conditional profiles, not paired causal changes on an identical successful-positive cohort.

C.8 Effects of the Joint Output Protocol

Requesting a description and support judgment alongside a box changes both rejection and localization behavior. Joint grounding lowers FGR in nine models and increases it in four; all-positive mIoU falls in twelve models. Tables A15 and A16 retain the full positive and negative results. LENS lowers FGR from 94.20% to 28.55%, while positive success falls from 49.6% to 25.6% and mIoU from 0.4874 to 0.2520. Seg-Zero lowers FGR from 97.10% to 38.00%, with positive success 55.0% to 46.4% and mIoU 0.5222 to 0.4549. LLaVA-OV instead increases FGR from 33.65% to 93.20%, while positive success falls from 37.2% to 16.2%; its joint relation CBR is 87.7%. Across all 13 models, joint FGR averages 48.25%, positive success 35.60%, and mIoU 0.3639. UniVG-R1 and Visual-RFT have joint FGR 87.40/65.55%, success 49.4/41.4%, and mIoU 0.4820/0.4249. Protocol changes therefore affect both rejection and localization.

MLLM	Direct FGR	Joint FGR	Desc. err.
Qwen3-VL-8B	21.15	22.45	9.20
Qwen2.5-VL-7B	34.50	33.85	4.00
InternVL3.5-8B	61.25	39.85	6.40
LLaVA-OV-7B	33.65	93.20	7.30
LENS-3B	94.20	28.55	55.60
Orsta-7B	44.15	37.55	7.05
Seg-R1-7B	61.75	31.60	5.90
Seg-Zero-7B	97.10	38.00	14.55
TreeVGR-7B	75.60	59.50	0.70
UniVG-R1-7B	99.80	87.40	0.45
Vision-R1-7B	37.80	43.80	5.70
VisionReasoner-7B	80.85	45.95	12.55
Visual-RFT-7B	28.25	65.55	1.50

Table A15: Unfiltered region emission and joint-description error (%). All three rates use 4,000 negative queries per model. Description error scores the generated joint-task text; FGR counts parsed coordinate emissions. Positive grounding quality appears in Table A16.

MLLM	Direct Eligible (%)	Joint Eligible (%)	Direct mIoU	Joint mIoU
Qwen3-VL-8B	47.6	46.4	0.4648	0.4585
Qwen2.5-VL-7B	48.4	42.4	0.4672	0.4223
InternVL3.5-8B	46.8	30.0	0.4653	0.3193
LLaVA-OV-7B	37.2	16.2	0.3351	0.2201
LENS-3B	49.6	25.6	0.4874	0.2520
Orsta-7B	30.2	27.8	0.3169	0.2948
Seg-R1-7B	52.2	41.0	0.4976	0.4059
Seg-Zero-7B	55.0	46.4	0.5222	0.4549
TreeVGR-7B	26.2	25.2	0.2811	0.2749
UniVG-R1-7B	50.8	49.4	0.4936	0.4820
Vision-R1-7B	29.6	29.6	0.3098	0.3046
VisionReasoner-7B	50.6	41.4	0.4959	0.4167
Visual-RFT-7B	42.2	41.4	0.4114	0.4249

Table A16: Original positive-correct coverage (Eligible, %) and all-positive mIoU under both grounding protocols. Description results appear in Table A10. Both protocols cover all 13 models.

Joint-description error uses 4,000 evaluated negative queries per model. Qwen3-VL, UniVG-R1, and Visual-RFT have error rates of 9.20%, 0.45%,

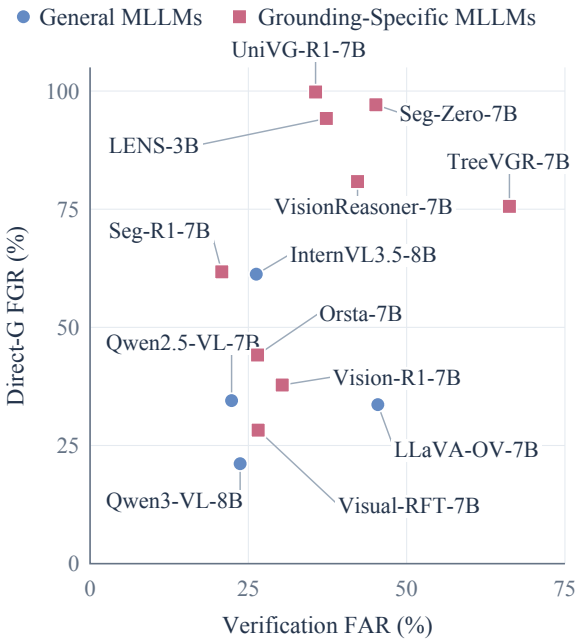


Figure A5: **Language acceptance and region emission.** Aggregate verification FAR (horizontal) and direct-grounding FGR (vertical) on negative queries. Each point is one MLLM; circles denote General MLLMs and squares denote Grounding-Specific MLLMs. Query-matched disagreement is reported separately in Appendix C.9.

and 1.50%. LENS has 55.60%, versus 4.20% object error in query-free captions. These measures have different tasks and denominators, so the percentages should not be interpreted as a shared error event.

C.9 Language–Region Disagreement

Differences between aggregate verification and grounding rates must be distinguished from disagreements on matched queries. The aggregate positive rate gap is

$$G = 100 \max\{0, \mathrm{FGR}_{\mathrm{DG}} - \mathrm{FA}_{\mathrm{verify}}\}. \quad (8)$$

Here rates are fractions, and verification FAR uses valid judgments. The gap is 35.00 points for InternVL3.5 ($61.25 - 26.25$) and 56.88 for LENS ($94.20 - 37.32$). G summarizes an aggregate rate difference with task-specific denominators. Figure A5 displays these aggregate rates for each model; the query-matched analysis below measures rejection and emission on identical requests.

Query-matched responses. All 13 MLLMs contribute 4,000 matched negative requests each (52,000 records; 1,000 source images). Matching requires identical model, query identifier, and query text. Of these records, 12,884 pair explicit verification rejection with direct-grounding emission (24.78%): 25.69% for object, 26.38% for association, 25.97% for attribute, and 21.06% for relation edits. The complete cohort includes Qwen3-VL's verification responses recovered from its repaired evaluation run. No model is excluded from this complete matched-query cohort. The matched-retention cohort below has a different eligibility rule (Table A20).

Models	Matched	No + box	Abstain
InternVL3.5-8B	2,000	700	0
LENS-3B	2,000	1,140	9
Orsta-7B	2,000	354	0
Qwen3-VL-8B	2,000	30	0
Seg-R1-7B	2,000	819	0
Seg-Zero-7B	2,000	1,031	15
TreeVGR-7B	2,000	172	617
UniVG-R1-7B	2,000	833	700
Vision-R1-7B	2,000	157	0
VisionReasoner-7B	2,000	784	2
LLaVA-OV-7B	2,000	10	0
Qwen2.5-VL-7B	2,000	243	0
Visual-RFT-7B	2,000	169	0
Total	26,000	6,442	1,343

Table A17: Matched negative-query records on the archived 500-image evaluation subset. "No + box" pairs an explicit verification rejection with direct-grounding emission. Abstention or invalid verification judgments stay in the full matched denominator and are not counted as explicit rejection.

D Verifier Training and Mitigation Analysis

We evaluate shared candidate support and structural comparison under a fixed-candidate protocol, reporting both their rejection gains and positive-grounding costs. The ASV settings and stage-wise mitigation results use the same evaluated candidates as the main paper. Appendix D.14 separately documents an earlier GRPO training study.

D.1 Evaluation Protocol

Mitigation and ablation metrics use the three-run inference averages defined in Appendix C. Unfiltered predictions, shared support, and full Adaptive Structured Verification (ASV) are compared on the same 13 upstream MLLMs. Each policy fixes the upstream candidate boxes, the original positive reference predictions, and the originally eligible positive groups. A retained prediction preserves its emission state and coordinates;

a rejected prediction becomes empty. Empty upstream predictions remain empty. Consequently, mitigation evaluates rejection of existing assignments rather than proposal generation or coordinate refinement.

Each model retains 1,000 positive and 4,000 negative queries, including 1,000 negatives per edit type. FGR counts parsed coordinate emissions after model-specific coordinate conversion, using the full negative-query denominator. CBR fixes each model's original positive-correct groups and positive boxes; negative refusals remain in its denominator. All-positive mIoU assigns zero to empty, rejected, or invalid positive predictions. Aggregate results weight models equally, including the mean of per-model CBR ratios. Table 4 gives the main operating points; the stage-wise and per-model tables retain this evaluation contract.

D.2 Verifier Adaptation and Calibration

JEV-2B adapts the TypeSafe/Open-Jev scoring interface on Qwen3.5-2B (Almeida, n.d.; Cai, 2026; Qwen Team, 2026). Frozen OmniVerifier-7B provides a second, image-conditioned score (Zhang et al., 2026); all upstream MLLMs also remain frozen. JEV reads the final prompt representation with a scalar head, whereas OmniVerifier uses the true-minus-false token-logit difference. Thus the shared-support input combines two readout interfaces for the same image, expression, and candidate region.

JEV initializes new rank-8 LoRA adapters with scale 16 on Qwen3.5-2B (Hu et al., 2022). Adapters cover both full-attention projections (`q/k/v/o_proj`) and linear-attention projections (`in_proj_qkv`, `out_proj`). A float32 scalar head reads the final 2,048-dimensional prompt state. Its training objective is

$$\mathcal{L} = \mathcal{L}_{\mathrm{NLL}} + 0.1\mathcal{L}_{\mathrm{Brier}}, \tag{9}$$

where the Brier component penalizes squared probability error (Brier, 1950). The adaptation dataset contains 2,000 COCO images disjoint from the 1,000 evaluation images and 16,000 candidate–expression records, balanced between KEEP and REJECT. Training shuffles records with seed 63, reserves the first 512 for temperature fitting, and trains on the remaining 15,488. Training runs in parallel on eight NVIDIA GeForce RTX 4090 GPUs for 1,000 optimizer updates, with an effective global batch size of 16 records per update. The adapter learning rate is 5×10^{-5} and the head learning rate is 10^{-4}. AdamW uses zero weight decay; gradient norms are clipped at 1.0. LoRA contributes 2,506,752 trainable parameters and the scalar head adds 2,049, totaling 2,508,801 parameters (approximately 0.113% of the 2.22B verifier).

Optimization dynamics. Figure A6 plots all 40 logged NLL and Brier points, each averaged over 25 optimizer updates. Mean NLL decreases from 0.3239 over the first 100 updates to 0.1355 over the last 100; mean Brier decreases from 0.1002 to 0.0317. The run processes 16,000 record presentations, approximately 1.033 passes over the training split. The fixed-order loop completes its first pass at update 968; the final 25-update window therefore contains revisited training records. The traces summarize training-set optimization.

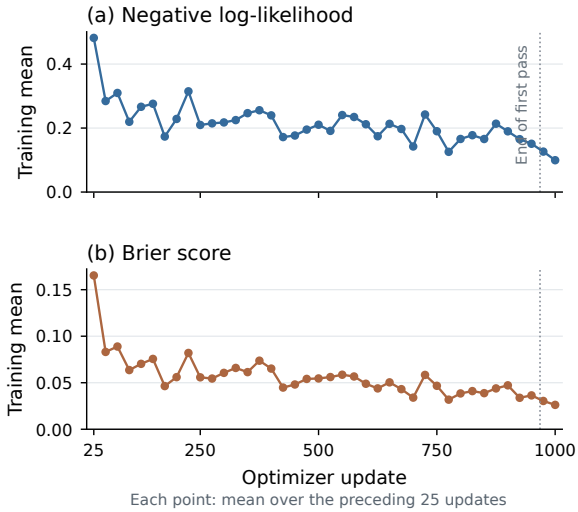


Figure A6: Training dynamics of the scalar-head JEV used by ASV. Each of the 40 points averages the preceding 25 optimizer updates; values retain the logged precision. The dotted line marks completion of the first data pass at update 968. These are training-set measurements.

D.3 Decision Heads and Predicate Routing

For direct grounding, feature order is $(s_{\mathrm{O},0}, s_{\mathrm{J},0})$ for the support head f_0 and $(s_{\mathrm{O},0}, s_{\mathrm{J},0}, \delta_{\mathrm{O}}, \delta_{\mathrm{J}})$ for the structural head f_1. Each upstream model and evaluation fold has separate logistic heads. The direct head estimates whether an expression is supported, matching the verifier adaptation labels; the joint head additionally incorporates candidate correctness to score the composed support–box output (Appendix D.12). Fitting uses positive/negative expression labels on usable rows;

only threshold selection restricts positives to originally correct candidates. Features use training-fold means and standard deviations, with unit scale for constant features. The unweighted binary logistic objective uses an L_2 coefficient of 1.0 on feature weights and an unpenalized intercept. Optimization starts at zero and uses up to 300 Newton updates, stopping when the largest damped update is below 10^{-9}.

The probe builder uses case-insensitive word-boundary matching with longer predicates first; multiple matches disable probe construction. The final policy independently identifies the longest recognized substring in the lowercased query. Structural comparison applies to *behind*, *in front of*, *under*, *on top of*, *above*, and *below*, provided both original scores, both alternative-score maxima, and a fitted structural head are available. Replacement expressions preserve the object mentions, exclude the original expression, and are deduplicated. Outside this scope, shared support supplies the final decision. Inside it, the final decision is the conjunction of support and structural decisions. Appendix D.13 gives the complete replacement mapping, extraction inventory, and checkpoint identities.

Each head requires at least 30 usable fitting rows and both labels. An unavailable support head or missing original scores preserves an emitted upstream candidate. An unavailable structural head, unmatched opposed predicate, or missing competitive scores falls back to shared support. These rules preserve $k_i \leq k_i^{(0)}$: the full keep set is a subset of the support keep set, itself a subset of the upstream candidates. With fixed eligibility and positive reference boxes, FGR and CBR therefore cannot increase relative to shared support. We report FGR and CBR jointly with correct-grounding retention and all-positive mIoU.

D.4 Cross-Fitted Threshold Selection

Direct-grounding evaluation uses image-level two-fold cross-fitting, assigning membership by MD5(image basename) mod 2. All paired queries from an image stay together. For each evaluation image, acceptance heads and thresholds are fitted on the other fold; head fitting and threshold selection share that training fold. The common nominal retention target is 0.95 on originally correct positives.

Let N_c denote originally correct calibration positives, U unrouted correct positives retained by support, and V_0 routed correct positives also retained by support. For $N_c > 0$, fixing the support decision constrains the retention attainable by changing η_1 to

$$\frac{U}{N_c} \leq \widehat{R}_{\mathcal{C}}(\eta_1) \leq \frac{U + V_0}{N_c}. \tag{10}$$

Finite samples and tied scores further discretize attainable values. The implementation selects η_0 by an order statistic of eligible training-fold positive support scores. To select η_1, it subtracts retained unrouted positives from the desired total, clips the required routed count to the number of covered positives, and takes the corresponding descending structural-score order statistic. A zero required count sets $\eta_1 = +\infty$; ties are retained through $\geq$. This structural threshold is selected before conjunction with $k_i^{(0)}$, whereas achieved retention is measured after the final conjunction. Nominal and achieved retention therefore need not coincide.

D.5 Support and Structural Ablations

Verifier-score complementarity. Table 3 compares one-score and two-score support heads with candidate availability, image folds, calibration target, and fallback rules fixed. Combining both scores lowers FGR to 22.55%, compared with 26.89% for OmniVerifier alone and 25.01% for JEV alone, while retaining 94.26% of correctly grounded positives. Both-score FGR is lower than either single-score configuration in all 13 models.

Structural increment. Table A18 reports the stage ablation on the same 13 models. Shared support lowers mean FGR from 59.23% to 22.55%; adding structural comparison lowers it to 19.03%. These are reductions of 36.68 and 3.52 percentage points, respectively, totaling 40.20 points (67.87% relative to the unfiltered mean). Correct-grounding retention is 94.26% for support and 91.83% for full ASV. All-positive mIoU changes from 0.4268 unfiltered to 0.4011 with support and 0.3914 with ASV. The structural stage thus adds rejection at a cost of 2.43 retention points and 0.0097 mIoU.

The largest incremental changes concern relational errors: ROH FGR falls from 32.89% to 26.36% (6.53 points), compared with EOH FGR from 12.20% to 11.71% (0.49 points). Relation CBR falls from 27.14% to 19.05% (8.09 points); association and attribute CBR fall by 0.18 and 0.24 points, while object CBR remains 1.69% at reported precision. Both semantic-level FGR means

Measure	Before	Support	Full	Δ
FGR	59.23	22.55	19.03	-3.52
EOH FGR	48.11	12.20	11.71	-0.49
ROH FGR	70.36	32.89	26.36	-6.53
Object CBR	16.47	1.69	1.69	+0.00
Assoc. CBR	25.63	9.22	9.04	-0.18
Attribute CBR	33.96	8.32	8.08	-0.24
Relation CBR	42.79	27.14	19.05	-8.09
R_{correct}	100.00	94.26	91.83	-2.43

Table A18: Equal-model means for the 13-model mitigation evaluation. The final column is full minus shared support in percentage points. EOH/ROH are FGR strata; the four individual edit columns use CBR.

and all four CBR means improve relative to unfiltered predictions.

Figure A7 summarizes the same stage-wise CBR changes. Read with the unfiltered profiles in Figure A4, it shows where rejection concentrates: shared support reduces all four error types, while the added structural stage mainly reduces relation reuse.

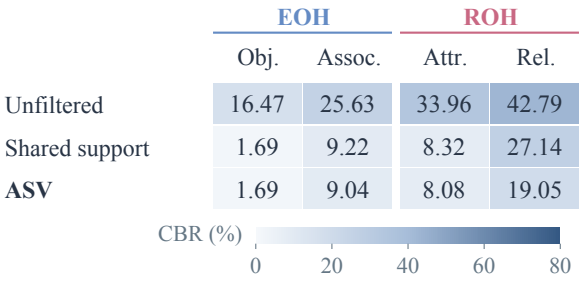


Figure A7: Four-type CBR through the two mitigation stages. Cells show equal-model means over 13 MLLMs with fixed candidates and eligibility. The shared 0–80% scale also applies to Figures A4 and A8. Retention costs appear in Table A18.

D.6 Matched-Retention Analysis

The matched-retention cohort includes all evaluated models except TreeVGR. For each model, shared support at nominal target 0.95 supplies the achieved-retention anchor. The full-policy curve uses targets $\{0.80, 0.85, 0.88, 0.90, 0.92, 0.94, 0.95, 0.96, 0.97, 0.98, 0.99\}$. Linear interpolation is permitted only inside its observed achieved-retention range. TreeVGR's 94.66% anchor exceeds its 93.13% full-policy maximum, so it is excluded without extrapolation. Table A20 lists every model's anchor and observed range. The artifact additionally records the two bracketing points and interpolation coefficient for each metric.

Within each metric, curve points are sorted by achieved retention, breaking ties by metric value, and adjacent bracketing points are linearly interpolated. The bracketing operating points can therefore differ by metric. The model-bootstrap intervals in Table A19 resample 12 paired model differences. The source-image bootstrap draws 1,000 image groups with replacement 10,000 times (seed 20261013), applying the same draw to every model and policy. Each draw recomputes counts and eligible denominators before equal-model averaging. Fitted heads, thresholds, original positive eligibility, interpolation anchors, and interpolation coefficients remain fixed. These intervals describe image-sampling variation conditional on the fitted policies.

Metric	Support	Full	Δ	Image CI
Rel. FGR	47.95	40.82	−7.13	[−9.28, −5.05]
Rel. CBR	28.76	23.00	−5.76	[−8.22, −3.49]

Table A19: Matched-retention results over 12 models. Rates are percentages; differences and 95% intervals are percentage points. Paired model-bootstrap intervals are $[-8.54, -5.73]$ for FGR and $[-7.23, -4.17]$ for CBR. Image intervals use grouped source-image resampling.

Models	Anchor	Full range	Included
InternVL3.5-8B	94.87	72.22–97.44	Yes
LENS-3B	93.95	76.21–95.97	Yes
Orsta-7B	94.04	76.16–94.04	Yes
Qwen3-VL-8B	94.54	74.79–95.80	Yes
Seg-R1-7B	95.40	75.10–95.79	Yes
Seg-Zero-7B	93.82	76.36–95.64	Yes
TreeVGR-7B	94.66	73.28–93.13	No
UniVG-R1-7B	94.49	75.20–98.82	Yes
Vision-R1-7B	93.24	75.68–95.95	Yes
VisionReasoner-7B	94.47	76.68–97.63	Yes
LLaVA-OV-7B	93.55	74.73–98.39	Yes
Qwen2.5-VL-7B	95.04	76.03–95.87	Yes
Visual-RFT-7B	93.36	74.41–95.73	Yes

Table A20: Matched-retention eligibility. Values are achieved retention (%). All 13 models enter query matching; 12 have anchors within the full-policy curve range.

D.6.1 Serial Threshold Calibration

The serial sensitivity probe preserves the all-eligible-positive target count but selects the structural-score order statistic only among support-retained routed positives, using the same image folds and final conjunction. Across the same 13 models, it attains FGR 20.04%, versus 22.55% for shared support, with correct retention 92.34% versus 94.26%. The two policies are evaluated at their respective achieved retentions.

D.7 Ordinary Expression-Verification Gate

The ordinary gate reuses each upstream model's image–expression verification response on its fixed direct-grounding candidate. An explicit, valid unsupported judgment rejects the candidate; support, abstention, or an unavailable judgment preserves it. No threshold is fitted to this binary baseline. The specialized heads use the same candidates, original positive-correct set, image folds, and nominal target 0.95. All 13 models have 1,000 positive and 4,000 negative requests; 43,170 candidates are emitted in total.

Table A21 reports equal-model means. Relative to the ordinary gate, the two-score head lowers FGR by 11.91 points and increases correct-grounding retention by 0.69 points. Incorrect rejection of originally correct positives is 6.43% for the ordinary gate and 5.74% for the two-score head; full ASV rejects 8.17% while attaining the lowest FGR. Error rejection uses each model's initially emitted negative candidates as its denominator.

Policy	FGR ↓	Err. rej. ↑	Ret. ↑	Calls
Unfiltered	59.23	0.00	100.00	0.00
Expression gate	34.46	36.40	93.57	1.00
Omni only	26.89	47.67	94.00	1.00
JEV only	25.01	52.29	93.30	1.00
Both scores	22.55	55.94	94.26	2.00
Full ASV	19.03	62.55	91.83	3.27

Table A21: Ordinary gating and dedicated verification on identical candidates. FGR uses all negative queries; error rejection (Err. rej.) uses initially emitted negative candidates; retention (Ret.) uses originally correct positives. Rates are percentages and costs are mean calls per emitted candidate.

Cost counts additional model/scorer invocations per emitted candidate: one upstream verification call for the ordinary gate, one for a single verifier, two for shared support, and 3.27 averaged over models for full ASV including collected rival scores. Across all emitted candidates, the corresponding totals are 43,170, 43,170, 86,340, and 140,868. These are call counts across different model interfaces; measured end-to-end verifier latency is not available in the saved runs.

D.8 Candidate-Box Dependence

We fix the image and supported expression and change only the candidate box in 40 distinct images. The correct box is the benchmark GT (IoU ≥ 0.95 with its COCO instance annotation); the replacement is a real same-class instance with target IoU < 0.1 and area above 225 pixels. Before scoring, the alternatives are visually screened by an assistant to violate the expression's attribute, relation, or participant role. This is a selected diagnostic probe, not an independently human-labeled accuracy sample. Both original and competitive expressions are rescored using the unchanged verifier interfaces.

JEV assigns lower scores to 37/40 alternative regions and Omni to 40/40 (Table A22). We apply the InternVL and UniVG decision heads fitted on each image's opposite benchmark fold; the 40 probe pairs are not added to fitting or calibration. The prediction interface exactly reproduces both original policies on each model's 5,000 saved requests. Table A23 then changes only the input box. Eight queries activate the opposed-predicate route; the others inherit shared support. Intervals use 10,000 paired image-bootstrap draws (seed 42).

Scorer	GT	Wrong instance	Paired drop [95% CI]
JEV	-0.79	-5.19	$4.40\ [3.22, 5.68]$
Omni	6.27	-4.10	$10.36\ [8.63, 12.07]$

Table A22: Fixed-image, fixed-expression box intervention: raw score means over 40 pairs. Scorers have different scales.

Head	Policy	GT kept	Wrong kept
InternVL	Support	40/40	16/40
	Full ASV	40/40	15/40
UniVG	Support	39/40	16/40
	Full ASV	39/40	15/40

Table A23: Policy dependence on the candidate box. Full-policy paired keep-rate drops are 62.5 points [47.5,77.5] for the InternVL head and 60.0 [45.0,75.0] for UniVG (95% image-bootstrap intervals).

D.9 Evidence Probes and Computational Cost

Table A24 specifies the image-swap and predicate-centering probes. Image swaps produce a mean OmniVerifier logit shift of 11.41 across 13 MLLMs, indicating sensitivity to image evidence with the query and candidate coordinates held fixed. On probe-covered candidates, subtracting the model-specific predicate mean from OmniVerifier's negative contrast $-\delta_{\mathrm{O}}$ yields mean expression-validity AUROC 0.598, compared with 0.613 before centering. Mean predicate-identity R^2 is 0.080 across 13 models. The centered AUROC quantifies expression-

validity discrimination after removing predicate means.

Probe	Protocol and aggregation
Image swap	Keep the query and candidate box coordinates fixed; score the original and mismatched images. Average paired logit shifts within each model, then across 13 models.
Predicate centering	Subtract each model's predicate mean from $-\delta_O$ on covered candidates. Use expression-validity labels for AUROC; average per-model AUROC and the predicate-explained variance fraction R^2.

Table A24: Evidence-probe protocols. Each upstream model receives equal weight in the aggregate probe statistics.

For a non-empty candidate, two original scores and K alternatives per verifier require $2 + 2K$ scorer–expression evaluations; support alone requires two. The CPU policy evaluation consumes previously collected scores. The predicate-and-score-availability rule selects 20.97% of all matched requests; additional OmniVerifier rival scores amount to 41.94% of its original support-score evaluation budget. Cost is measured in scorer–expression evaluations; the 41.94% figure covers OmniVerifier rival scores.

D.10 Per-Model Mitigation Results

Tables A25 and A26 retain all per-model results behind the 13-model means. Every model has lower FGR with full ASV than with shared support. The tables pair CBR with retention for each model's fixed candidates and eligible positives.

With structural comparison, Seg-Zero's relation CBR changes from 35.64% to 24.73%, with retention from 93.82% to 91.27%; LENS changes from 40.73% to 27.02% and from 93.95% to 91.94%, respectively. TreeVGR changes from 7.63% to 6.11% relation CBR and from 94.66% to 90.84% retention. Vision-R1 changes from 8.78% to 8.11% and from 93.24% to 91.89%. Qwen2.5-VL changes from 32.64% to 22.31% and from 95.04% to 91.74%. Figure A8 uses a common 0–80% CBR scale for these three models; the LENS and Seg-Zero profiles appear in Figure 4.

D.11 Qualitative Comparison across Eight Models

Figures A9–A12 compare 16 selected image–query cases across eight MLLMs, with four cases per semantic edit type. Each row fixes the image and query across models and pairs the upstream Direct-*G* candidate with its ASV decision. These examples extend Figure 5 with cross-model views of successful rejection and residual false acceptance.

	EOH		ROH	
	Obj.	Assoc.	Attr.	Rel.
TreeVGR-7B				
Unfiltered	16.03			
Shared support	0.00	2.29	2.29	7.63
ASV	0.00	2.29	0.76	6.11
Vision-R1-7B				
Unfiltered	14.19			
Shared support	0.00	4.05	0.68	8.78
ASV	0.00	4.05	0.68	8.11
Qwen2.5-VL-7B				
Unfiltered	44.63			
Shared support	2.07	9.50	10.33	32.64
ASV	2.07	9.50	10.33	22.31

CBR (%) 0 20 40 60 80

Figure A8: Three-policy CBR heatmaps for TreeVGR, Vision-R1, and Qwen2.5-VL, using fixed candidates and eligible positives with a common 0–80% scale. Tables A25 and A26 report the accompanying retention and all-positive mIoU.

Shared failure patterns and rejection. Different models often assign visible distractors to the same unsupported query. In the Object cases, the orange query attracts an apple region, while the tent queries attract umbrellas or nearby regions (Figure A9). The Attribute cases retain a visible target but attach an unsupported property: the vase is not red, the fork is not golden, and the sink is not wooden (Figure A11). All eight models return empty outputs after ASV for the four displayed Object cases and the four Attribute cases. This illustrates how candidate rejection can correct assignments that localize a visible object without satisfying the full expression.

Residual false acceptance. The Association and Relation panels also expose model-dependent failures. For the salt-shaker query in Figure A10, InternVL3.5 and LENS retain their original candidates, while the other six models reject them. The same two models retain the catcher candidate for the unsupported in-front-of relation in Figure A12. All eight models reject the other three displayed cases in each panel. Retained outputs preserve the original coordinates, making the remaining errors

MLLM	FGR	EOH	ROH	Obj.	Assoc.	Attr.	Rel.	R_{correct}	mIoU
Qwen3-VL-8B	14.00	6.60	21.40	1.26	7.98	8.40	24.37	94.54	0.4379
Qwen2.5-VL-7B	20.50	11.10	29.90	2.07	9.50	10.33	32.64	95.04	0.4435
InternVL3.5-8B	27.55	14.90	40.20	2.56	12.82	13.68	30.77	94.87	0.4420
LLaVA-OV-7B	20.10	10.00	30.20	2.69	10.75	7.53	30.11	93.55	0.3131
LENS-3B	27.80	16.00	39.60	3.23	16.13	13.31	40.73	93.95	0.4562
Orsta-7B	20.65	11.30	30.00	0.00	2.65	2.65	11.26	94.04	0.2936
Seg-R1-7B	24.25	13.40	35.10	2.68	10.73	11.11	35.63	95.40	0.4728
Seg-Zero-7B	26.30	14.80	37.80	2.55	12.36	11.64	35.64	93.82	0.4868
TreeVGR-7B	25.20	13.80	36.60	0.00	2.29	2.29	7.63	94.66	0.2652
UniVG-R1-7B	24.95	13.60	36.30	1.57	9.84	8.66	28.74	94.49	0.4642
Vision-R1-7B	21.05	11.20	30.90	0.00	4.05	0.68	8.78	93.24	0.2906
VisionReasoner-7B	25.40	13.80	37.00	2.37	12.25	12.65	37.55	94.47	0.4663
Visual-RFT-7B	15.35	8.10	22.60	0.95	8.53	5.21	28.91	93.36	0.3820

Table A25: Shared support verification: per-model results corresponding to the 13-model means in Table 4. EOH/ROH are FGR strata and the four edit columns report CBR. Rates are percentages; mIoU is in 0–1. The eligible positive set is fixed within each model.

MLLM	FGR	EOH	ROH	Obj.	Assoc.	Attr.	Rel.	R_{correct}	mIoU
Qwen3-VL-8B	12.45	6.50	18.40	1.26	7.98	8.40	17.65	91.60	0.4246
Qwen2.5-VL-7B	17.40	10.70	24.10	2.07	9.50	10.33	22.31	91.74	0.4281
InternVL3.5-8B	23.95	14.80	33.10	2.56	12.82	13.68	21.79	92.74	0.4326
LLaVA-OV-7B	16.80	9.60	24.00	2.69	10.75	7.53	22.04	91.40	0.3066
LENS-3B	23.05	15.50	30.60	3.23	15.32	12.10	27.02	91.94	0.4464
Orsta-7B	16.95	11.00	22.90	0.00	2.65	2.65	7.28	90.73	0.2844
Seg-R1-7B	20.20	12.90	27.50	2.68	10.73	11.11	22.99	93.10	0.4615
Seg-Zero-7B	21.90	13.60	30.20	2.55	11.64	11.64	24.73	91.27	0.4738
TreeVGR-7B	19.45	12.60	26.30	0.00	2.29	0.76	6.11	90.84	0.2559
UniVG-R1-7B	21.60	12.70	30.50	1.57	9.06	8.66	20.87	92.13	0.4537
Vision-R1-7B	18.05	10.90	25.20	0.00	4.05	0.68	8.11	91.89	0.2874
VisionReasoner-7B	22.10	13.40	30.80	2.37	12.25	12.25	26.88	92.49	0.4566
Visual-RFT-7B	13.55	8.00	19.10	0.95	8.53	5.21	19.91	91.94	0.3766

Table A26: Adaptive Structured Verification: per-model results corresponding to the 13-model means in Table 4. EOH/ROH are FGR strata and the four edit columns report CBR. Rates are percentages; mIoU is in 0–1. The eligible positive set is fixed within each model.

failures of acceptance rather than box refinement. These selected negative examples complement the aggregate rejection and positive-retention results in Tables A25 and A26.

D.12 Joint-Grounding Extension

The joint extension evaluates all 13 models using a separately fitted configuration and scores collected on their joint-grounding candidates. UniVG-R1 and Visual-RFT normalized coordinates are converted to image pixels before scoring. The support head uses the OmniVerifier original score and the stored raw JEV scalar-head logit. The structural head adds only the OmniVerifier alternative-minus-original margin: three features for joint grounding, versus four for direct grounding. Both configurations use monotone conjunction and preserve the original box coordinates.

Joint head fitting uses candidate-correctness labels: a positive query receives label one only when upstream IoU is at least 0.5. Routing checks whether a collected rival surface belongs to the opposed-predicate set. Two-fold membership is determined by MD5 of the base image-group identifier, keeping paired queries together. Fitting and threshold selection use the opposite fold from evaluation; feature standardization uses only the fitting fold. Table A27 makes the task-specific differences explicit. Direct and joint grounding use task-specific feature and label configurations.

Positive-label control. Direct-G estimates expression support, while Joint-G estimates support with a correct candidate assignment. Both calibrate retention on originally correct positives. To isolate this label choice, we refit each task with both definitions while fixing its candidates, scores, feature configuration, image folds, and target 0.95 (Table A28). Switching Direct-G to candidate-

Task	Head-fitting label $y = 1$	Structural features
Direct	Positive expression	Two original scores and both verifier contrasts
Joint	Positive query with upstream IoU ≥ 0.5	Two original scores and OmniVerifier contrast

Table A27: Task-specific decision-head configurations. Other fitting rows have $y = 0$; retention thresholds use originally correct positives. Both tasks use image-grouped two-fold cross-fitted acceptance heads.

correct labels changes full FGR from 19.03% to 19.55%; switching Joint-G to expression-valid labels changes it from 18.45% to 18.85%. This comparison reports the effect of the implemented target definitions without changing the task's other settings.

Task	Head label	FGR ↓	Retention ↑
Direct-G	Expression valid	19.03	91.83
	Candidate correct	19.55	92.16
Joint-G	Expression valid	18.85	93.28
	Candidate correct	18.45	92.99

Table A28: Full-policy positive-label control on the same 13 models. Candidate correct means a supported expression with upstream IoU ≥ 0.5. All values are percentages; candidates, scores, features, image folds, and the 0.95 retention target stay fixed; heads and thresholds are refitted for each label definition.

At nominal target 0.95, joint eligibility fixes each model's original positives with IoU ≥ 0.5. The full 1,000-positive and 4,000-negative denominators, unchanged positive reference boxes, and retained negative refusals follow the contract above. Equal-model FGR is 48.25% unfiltered, 19.70% with shared support, and 18.45% with the full extension. Full ASV reduces FGR in all 13 models relative to both unfiltered predictions and shared support. All four mean CBR values decrease relative to unfiltered predictions; no per-model CBR increases. Table A29 reports mean CBR and positive-grounding outcomes, while Table A30 supplies the underlying per-model operating points.

The joint extension retains 92.99% of originally correct positives, compared with 93.95% for support; all-positive mIoU is 0.3341, compared with 0.3379 for support and 0.3639 unfiltered. Relation CBR decreases from 25.18% with support to 23.08% with full ASV. As in direct grounding, the reduction in false assignments accompanies a measurable positive-grounding cost.

Metric	Unfiltered	Support	ASV
FGR	48.25	19.70	18.45
Object CBR	12.30	1.54	1.54
Association CBR	21.62	9.01	8.79
Attribute CBR	27.20	8.43	7.75
Relation CBR	41.18	25.18	23.08
Correct retention	100.00	93.95	92.99
mIoU	0.3639	0.3379	0.3341

Table A29: Joint-grounding mitigation: equal-model means over 13 models at nominal target 0.95. Eligibility and positive reference boxes are fixed. Rates are percentages; mIoU includes all positives.

D.13 Executable Reproduction Details

Hardware and repeated inference. Training and inference use eight NVIDIA GeForce RTX 4090 GPUs. Each training run uses all eight GPUs in parallel. JEV adaptation uses an effective global batch size of 16 records per optimizer update. All reported evaluation metrics, including ablations, average three inference runs as specified in Appendix C.

Checkpoint and parser identity. Table A31 identifies all upstream checkpoints and the verifier bases. JEV uses the final adapter and scalar head from the supervised 1,000-update run, with temperature 1.776567; it does not use an earlier GRPO checkpoint. The adapter, scalar head, and temperature file have SHA-256 prefixes `1ae071150106`, `dc265b89b315`, and `d11ec5ce48cf7`, respectively. Full digests and the trained adapter are included in the reproduction bundle.

The source supplement includes each of the 13 models' full system and four task templates in `repro/model_task_templates.json`. The evaluator snapshot preserves chat adapters, image preprocessing, generation options, output wrappers, refusal handling, and model-specific coordinate parsers. Direct and joint policy snapshots accompany the candidate-source manifests and per-request decisions. Each rival record explicitly stores its scoring expression, candidate coordinates, and score; the scorer reads this expression verbatim. Thus both the exact input strings and parser precedence are specified beyond the representative prompts in Appendix C.2.

JEV input and adaptation labels. JEV receives the full, unmarked RGB image and absolute integer pixel coordinates in its text prompt; no crop is supplied. The final prompt-position representation feeds the scalar head rather than parsing a gener-

Shared support verification										
MLLM	Eligible (%)	FGR	EOH	ROH	Obj.	Assoc.	Attr.	Rel.	R_{correct}	mIoU
Qwen3-VL-8B	46.4	14.50	7.30	21.70	1.29	6.47	6.90	22.84	92.24	0.4220
Qwen2.5-VL-7B	42.4	17.85	9.50	26.20	2.36	9.43	11.32	27.83	93.87	0.3913
InternVL3.5-8B	30.0	21.25	11.80	30.70	1.33	8.67	7.33	18.00	94.67	0.2966
LLaVA-OV-7B	16.2	23.10	12.70	33.50	1.23	17.28	14.81	43.21	96.30	0.2056
LENS-3B	25.6	9.60	4.90	14.30	2.34	7.03	4.69	25.00	91.41	0.2269
Orsta-7B	27.8	16.65	8.70	24.60	0.00	6.47	5.04	12.23	95.68	0.2717
Seg-R1-7B	41.0	17.50	9.00	26.00	1.95	10.73	11.22	33.66	93.66	0.3766
Seg-Zero-7B	46.4	18.60	9.30	27.90	2.16	9.05	8.62	30.17	94.40	0.4245
TreeVGR-7B	25.2	22.20	11.80	32.60	0.00	3.17	3.17	6.35	94.44	0.2539
UniVG-R1-7B	49.4	26.95	15.30	38.60	2.02	11.34	10.53	32.39	92.71	0.4472
Vision-R1-7B	29.6	21.45	11.20	31.70	0.00	3.38	1.35	6.08	92.57	0.2830
VisionReasoner-7B	41.4	23.15	12.80	33.50	2.90	14.49	13.04	39.61	95.65	0.3950
Visual-RFT-7B	41.4	23.25	12.50	34.00	2.42	9.66	11.59	29.95	93.72	0.3986
ASV: joint-grounding extension										
MLLM	Eligible (%)	FGR	EOH	ROH	Obj.	Assoc.	Attr.	Rel.	R_{correct}	mIoU
Qwen3-VL-8B	46.4	13.30	6.80	19.80	1.29	6.47	6.47	20.26	90.09	0.4122
Qwen2.5-VL-7B	42.4	16.40	9.00	23.80	2.36	9.43	10.38	24.06	93.40	0.3893
InternVL3.5-8B	30.0	20.10	11.50	28.70	1.33	8.67	6.67	16.00	94.67	0.2960
LLaVA-OV-7B	16.2	20.30	10.90	29.70	1.23	14.81	12.35	38.27	92.59	0.1972
LENS-3B	25.6	9.25	4.90	13.60	2.34	7.03	4.69	23.44	91.41	0.2269
Orsta-7B	27.8	14.85	8.10	21.60	0.00	6.47	3.60	12.23	92.81	0.2598
Seg-R1-7B	41.0	17.10	8.90	25.30	1.95	10.73	11.22	32.20	93.66	0.3766
Seg-Zero-7B	46.4	18.20	9.30	27.10	2.16	9.05	8.62	28.02	94.40	0.4244
TreeVGR-7B	25.2	21.30	11.50	31.10	0.00	3.17	2.38	6.35	94.44	0.2532
UniVG-R1-7B	49.4	26.35	15.20	37.50	2.02	10.93	10.53	30.36	92.31	0.4450
Vision-R1-7B	29.6	18.80	10.20	27.40	0.00	3.38	0.68	6.08	91.22	0.2766
VisionReasoner-7B	41.4	21.95	12.30	31.60	2.90	14.49	12.08	36.23	94.20	0.3894
Visual-RFT-7B	41.4	21.95	12.00	31.90	2.42	9.66	11.11	26.57	93.72	0.3968

Table A30: Per-model joint-grounding extension results. The four edit columns report CBR; Eligible gives the original positive-correct proportion (%), fixed across policies and types. FGR counts parsed coordinate emissions after model-specific coordinate conversion, as specified in Appendix D.12. Rates are percentages; mIoU uses all positive queries.

Model checkpoint	Revision prefix
Qwen2.5-VL-7B	`cc594898137f`
LENS-3B	`1b4c57b667a7`
Visual-RFT-7B	`cc595b120835`
Seg-R1-7B	`dbbd49a93b7d`
Seg-Zero-7B	`7764ee846ff4`
VisionReasoner-7B	`d27f35b206b8`
TreeVGR-7B	`5c8b749d6ff9`
Orsta-7B	`14a1097ab21b`
Vision-R1-7B	`ae01b49ed9f1`
UniVG-R1-7B	`8c137311d12e`
Qwen3-VL-8B	`0c351dd01ed8`
LLaVA-OV-7B	`0d5068052768`
InternVL3.5-8B	`9bb6a56ad9cc`
OmniVerifier-7B	`f2d958383ae4`
JEV base (Qwen3.5-2B)	`15852e8c1636`

Table A31: Evaluated checkpoint identities. The source supplement supplies full 40-character revisions, configuration and tokenizer hashes, and local adaptation-file hashes in `repro/checkpoints.json`.

ated decision. Every adaptation image contributes four positive–negative pairs, each using the shared GT box: supported expressions receive KEEP and edited negatives receive REJECT. Training coordinates use integer truncation, `int(v)`; scoring coordinates use `int(round(v))`. An exhaustive check matches all 16,000 labels and boxes to the source pairs, with 8,000 KEEP and 8,000 REJECT records. Candidate IoU does not define these adaptation labels.

Table A32: **Exact JEV Scoring Prompt.**

Full image + absolute pixel coordinates

Inspect the image region [`{x0},{y0},{x1},{y1}`]. Determine whether the phrase "`{q}`" is fully true for the object in that exact region. Return exactly one JSON object with exactly these two fields: `{"binding":"MATCH" or "MISMATCH" or "NA", "decision":"KEEP" or "REJECT"}`. Do not explain. Do not use Markdown.

OmniVerifier input and score. The full image is marked with an RGB $(255, 0, 0)$ rectangle of

width $\max(2, \mathrm{round}(0.008 \min(W, H)))$. Images larger than 896 pixels on their longest side are resized proportionally with bicubic interpolation after drawing the box. No region crop is used. After applying the chat template, the assistant prefix `{"answer":` is appended. The score subtracts log-sum-exp over the first-token IDs of `false`, `False`, and their leading-space variants from the corresponding true-token log-sum-exp, after deduplicating token IDs.

Table A33: **Exact OmniVerifier Scoring Prompt.**

Full image + red candidate rectangle

The red rectangle in the image marks a candidate region.
Claim about the red-boxed region: "`{q}`".
Carefully analyze the image and determine whether the object, its attributes, and its spatial relationships stated in the claim are all correctly satisfied by the red-boxed region. If the claim is accurate, answer true; otherwise false.
Respond strictly in JSON: `{"answer": true/false}`

Complete probe construction. Table A34 gives the effective ordered replacements with at most two alternatives. Only the matched span is replaced; prefix, suffix, and object mentions are preserved. The regular expression recognizes the following surfaces, with longer alternatives first:

> on the right side of; on the left side of; standing next to; sitting next to; to the right of; to the left of; surrounded by; pointing at; in front of; attached to; standing on; looking at; underneath; covered by; leaning on; sitting on; on top of; far from; right of; touching; close to; carrying; drinking; lying on; holding; wearing; next to; left of; beneath; between; held by; worn by; inside; eating; behind; beside; riding; around; below; among; under; above; with; near; onto; into; over; on; in; at; by.

Multiple recognized predicate occurrences, an undefined semantic family, or an unmapped competitor rule yields no competitive probe. The ambiguity blocklist is {at, by, in, into, on, onto, over, with}. Equivalent paraphrases and weakened variants are disabled in this configuration. The full inventory, family lookup, rule ordering, and source snapshot accompany `repro/query_predicate_mapping.json`.

Policy folds and outputs. Direct grounding hashes the complete COCO image basename (including `.jpg`); joint grounding hashes the base image-group identifier before its task suffix. Both use MD5 modulo two, so paired queries stay together although the two tasks' fold assignments differ. Heads and thresholds for an evaluation image use only the opposite fold. The supplement records the head features, labels, fallback rules, standardization, and threshold order statistics, together with actual per-request policy outputs.

Matched predicate	Ordered replacements
in front of	behind, next to
behind	in front of, next to
above	below, next to
below	above, next to
on top of	under, next to
under	on top of, next to
to the left of	to the right of
to the right of	to the left of
left of	right of
right of	left of
next to	behind, in front of
beside	behind, in front of

Table A34: Complete effective $Q(q)$ replacement map with the default two-probe cap. The first six rows form the direct policy's opposed-predicate route; collecting other probes does not activate that route.

D.14 Exploratory GRPO Verifier Training

We additionally document an earlier generative-verifier experiment using GRPO (Shao et al., 2024) in ms-swift (Zhao et al., 2025). Two Qwen3.5-2B runs, R1 and R2, generate `binding` and `decision` fields from the same training set of 2,000 images and 16,000 candidate–expression records, balanced between KEEP and REJECT. These runs are separate from the scalar-head JEV adaptation in Appendix D.2; they do not produce the ASV results reported in the main tables. Table A35 reports their training configurations.

Reward configurations. R1 combines format, ROH binding, and decision components as $R_1 = 0.1r_F + r_B + r_D$. Its component values are $r_F = 0.1$ for a valid typed response and zero otherwise; $r_B = 1.5$ for a correct ROH binding and zero otherwise; and $r_D = 1$ for a correct decision, -0.2 for an unparseable decision, and zero for a parsed incorrect decision. Binding targets are derived from the ROH type and KEEP/REJECT label, not independently annotated process supervision.

R2 uses $R_2 = 0.1r'_F + r'_D$, omitting binding reward. Here $r'_F = 0.1$ for valid format and -0.2 otherwise; its decision component is evaluated from the parsed decision field. Both configurations therefore give valid format an effective contribution of 0.01. R1 and R2 compare complete reward configurations, including their parsers and invalid-output penalties.

Optimization dynamics. Figure A13 uses all 400 recorded steps from each run. Both runs'

smoothed rewards increase from their initial values; final logged KL is 0.02337 for R1 and 0.01336 for R2. The curves track each configuration's training reward on its own reward scale.

Training diagnostics. Training uses eight GPUs: per-device batch 1 and accumulation 8 yield 64 completions per update, corresponding to eight prompt records with eight samples each. Both runs end at epoch 0.2 after 400 updates. No logged completion is length-clipped; the gradient norm exceeds the configured clip threshold of 1.0 in 4/400 R1 steps and 6/400 R2 steps. Both runs use fixed 400-update schedules with evaluation disabled and without a validation split or validation-based checkpoint selection.

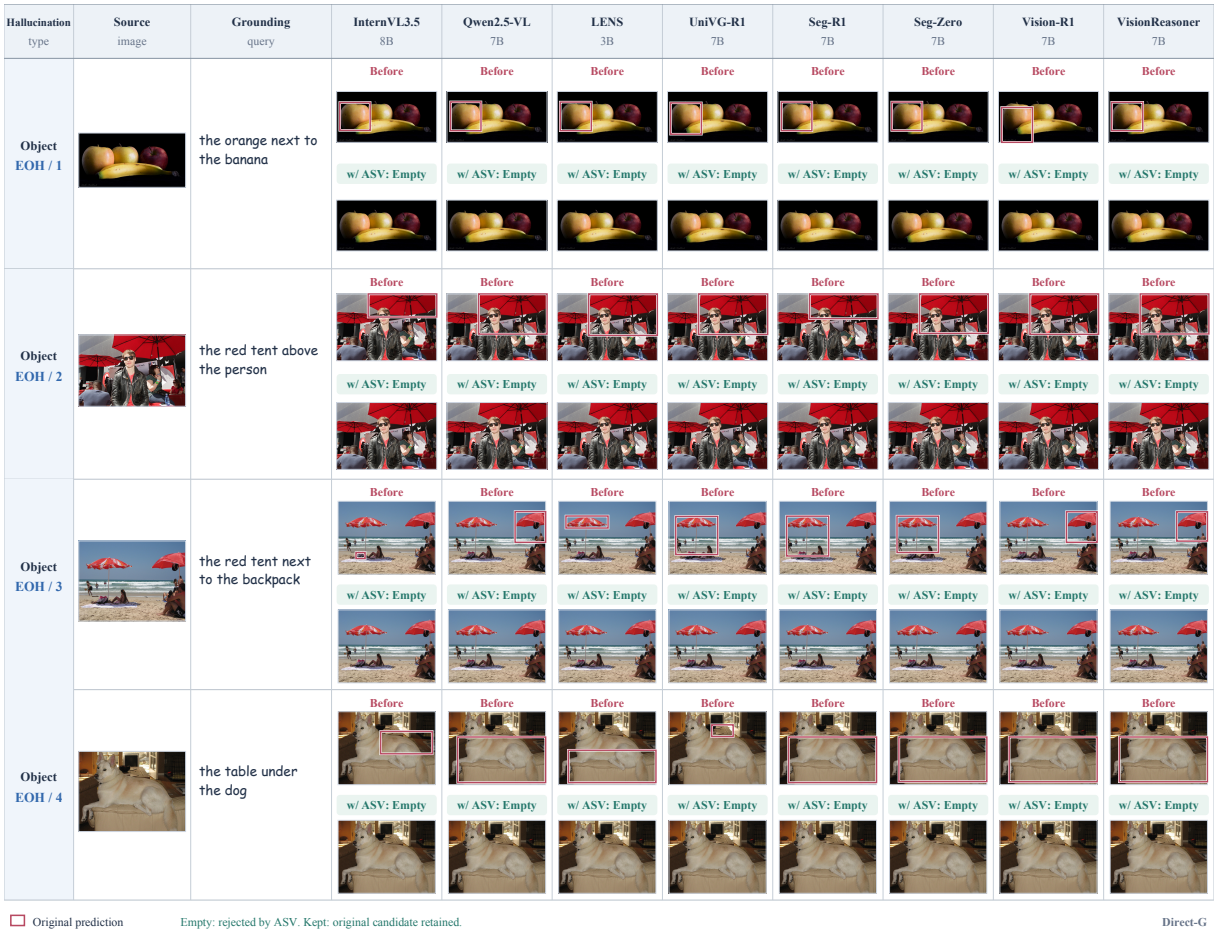


Figure A9: **Multi-model ASV comparison: Object (EOH).** Four selected Direct-*G* cases are evaluated across eight MLLMs. Each row shares the image and query across model columns. Red boxes mark upstream candidates; Empty denotes ASV rejection, and Kept denotes preservation of the original coordinates. All displayed Object candidates are rejected.

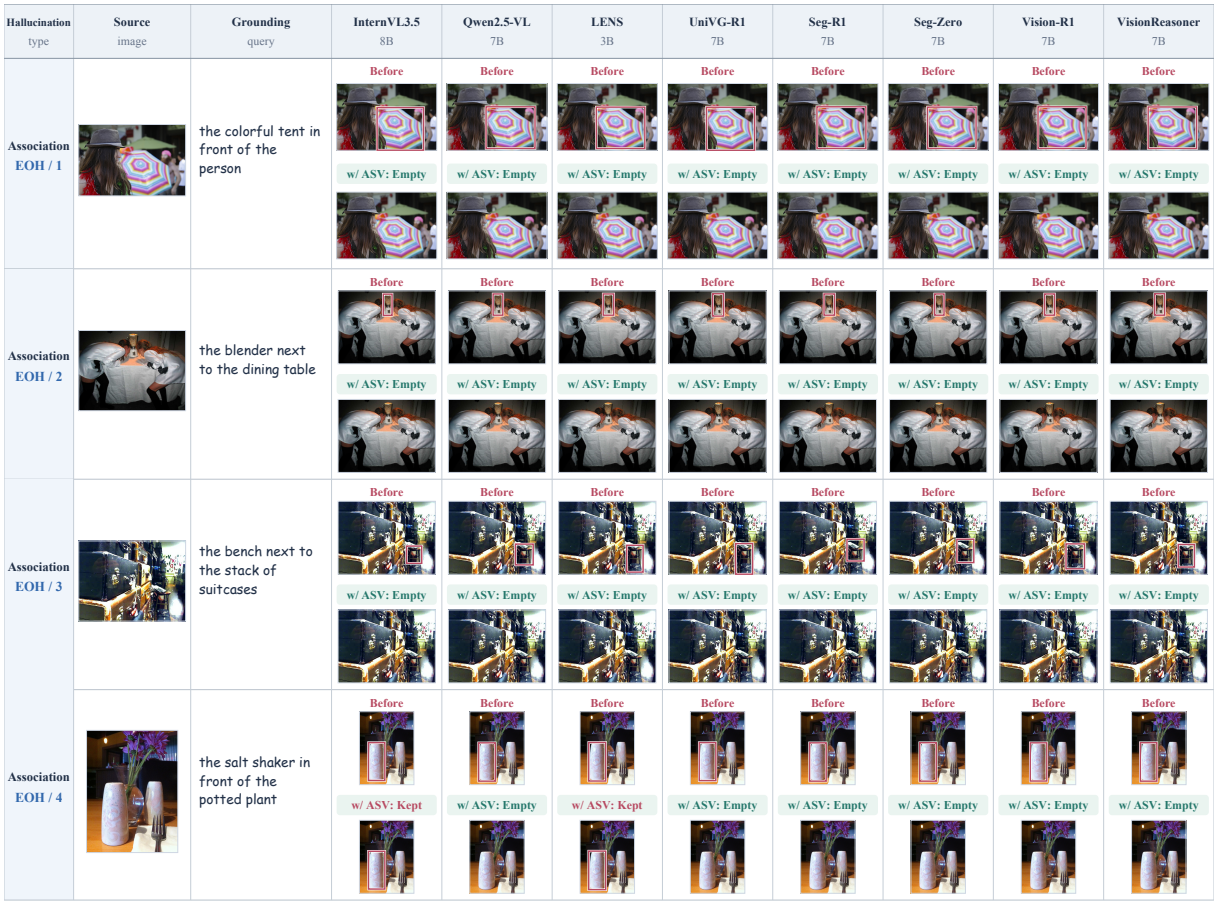


Figure A10: **Multi-model ASV comparison: Association (EOH).** Four selected cases use the model columns and output conventions of Figure A9. All models reject the first three queries. For the salt-shaker query, InternVL3.5 and LENS retain their candidates, while the other six models return empty outputs.

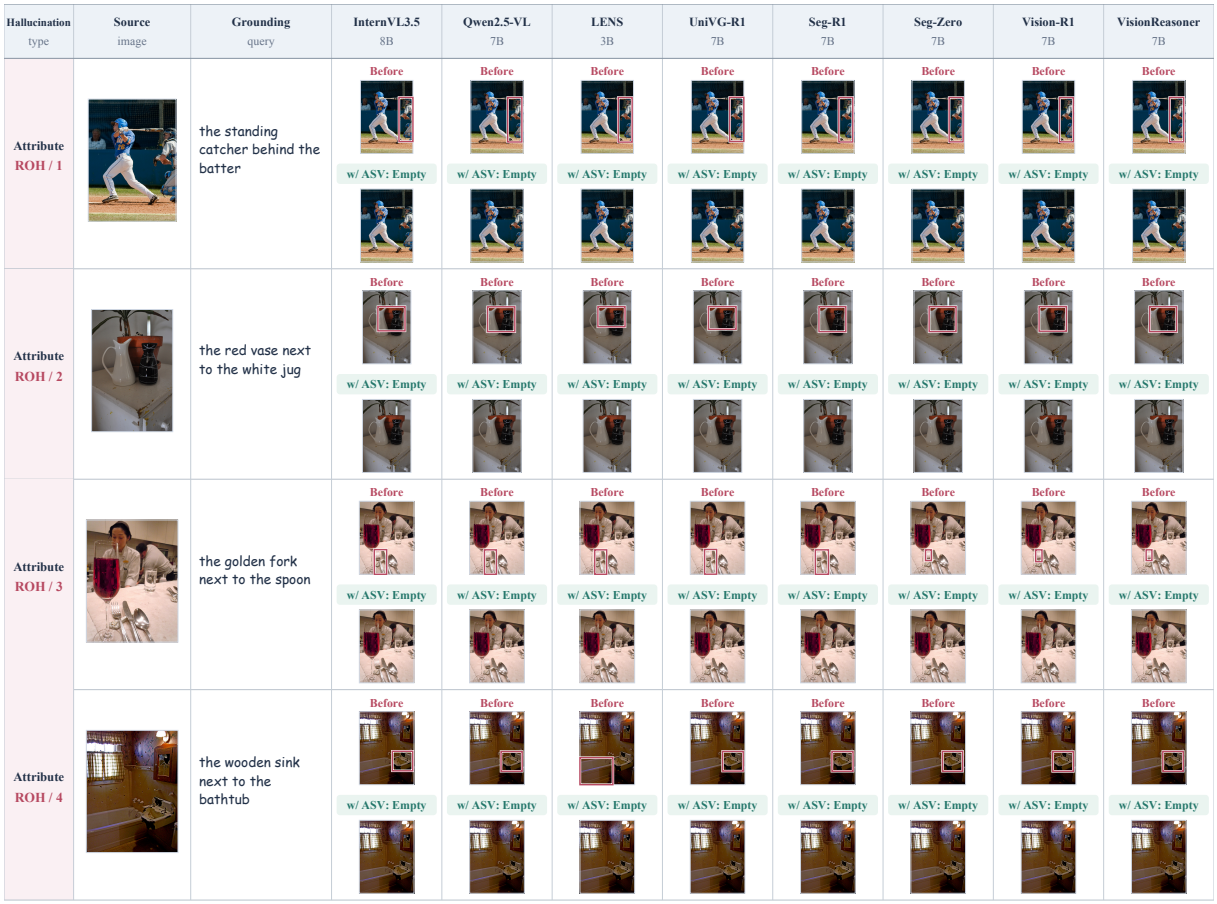


Figure A11: **Multi-model ASV comparison: Attribute (ROH).** Four selected cases challenge posture, color, and material properties of visible objects. The model columns and output conventions follow Figure A9. All eight models return empty outputs after ASV for each displayed query.

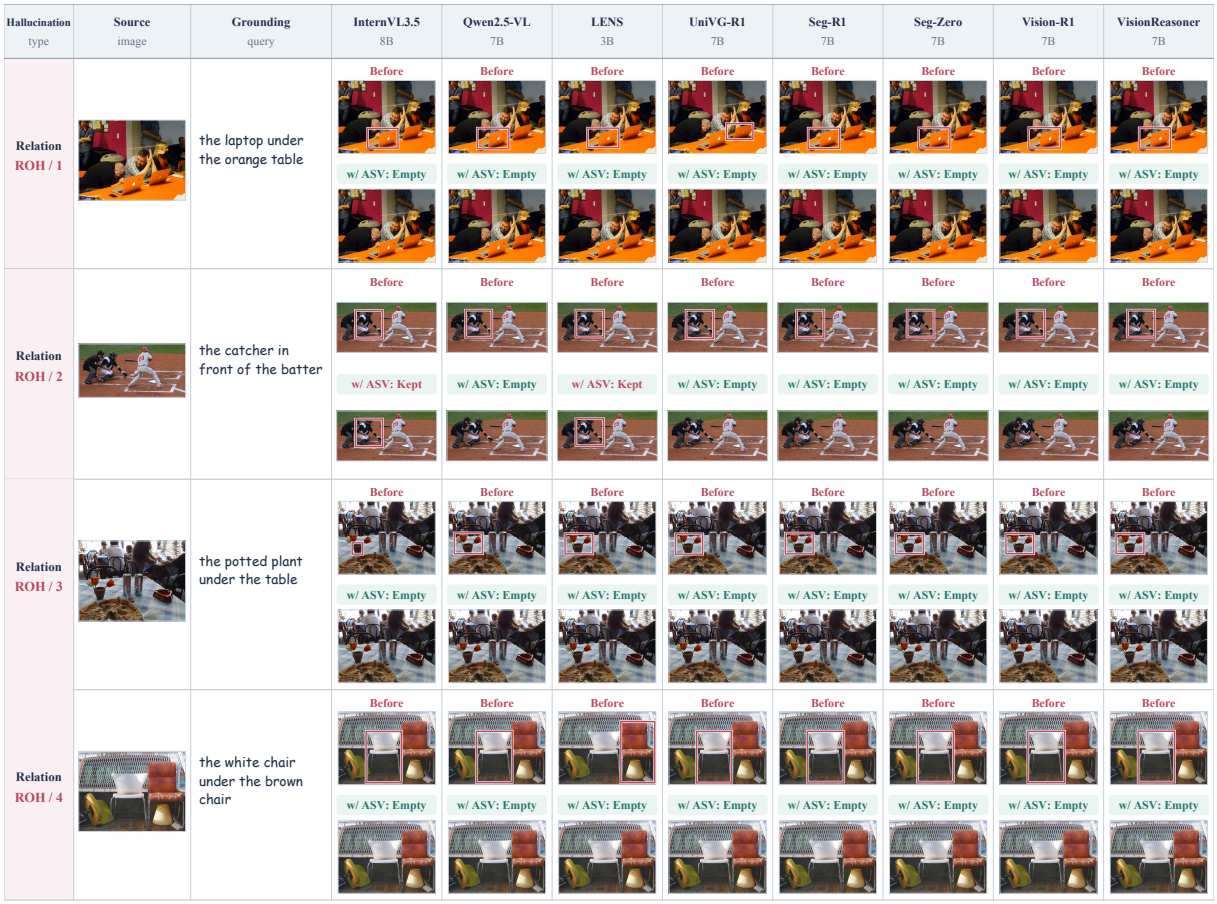


Figure A12: **Multi-model ASV comparison: Relation (ROH).** Four selected cases challenge spatial relations between visible objects, using the conventions of Figure A9. InternVL3.5 and LENS retain the catcher candidate for the in-front-of query; the other six models reject it. All models reject the other three queries.

Setting	R1 and R2
Base model	Qwen3.5-2B, revision `15852e8c`
Training data	2,000 images; 16,000 records (8,000 KEEP / 8,000 REJECT)
LoRA	Rank 32, scale 32, dropout 0.05
Target projections	`q_proj`, `k_proj`, `v_proj`, `o_proj`
Optimizer	Fused AdamW; LR 2×10^{-6}; weight decay 0.1
Adam parameters	$(\beta_1, \beta_2) = (0.9, 0.95)$; $\epsilon = 10^{-8}$
Schedule / updates	Cosine; warmup ratio 0.05; 400 steps
GPU count / per-device batch / accumulation	8 / 1 / 8
Generation batch / group size	64 completions / 8 per prompt
Effective update / data exposure	64 completions; 8 prompt records per update; 0.2 epoch
Sampling	Temperature 1.0; at most 64 new tokens; thinking disabled
GRPO controls	KL coefficient 0.01; clipping $\epsilon = 0.2$; group reward scaling
Precision / gradient controls	bfloat16; checkpointing enabled; norm clip 1.0
Randomness / data split	Model and data seeds 63; no automatic evaluation split

Table A35: Saved settings shared by the two historical GRPO runs. The generation batch is completion-level. The 400 updates correspond to 3,200 record presentations, not 25,600 distinct prompts. Both runs use the same historical verifier training file; these settings do not describe the JEV adaptation pool.

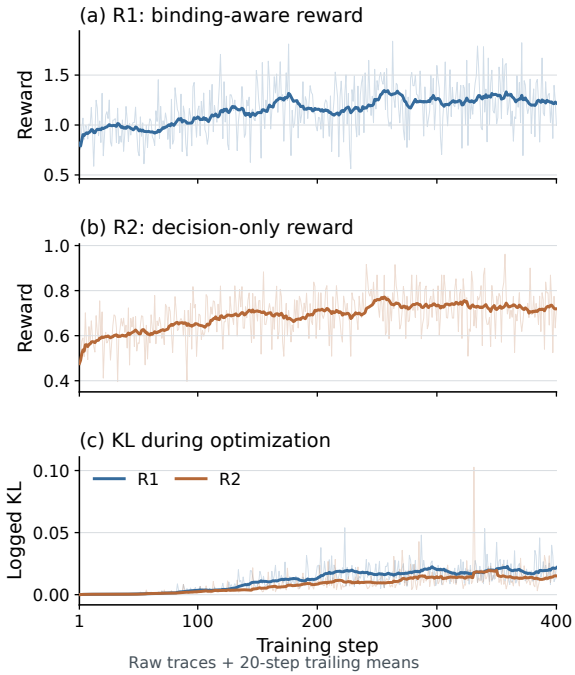


Figure A13: GRPO optimization over 400 steps. Faint traces retain every step; solid curves show trailing 20-step means. Reward panels use separate scales.

E Qualitative Cases and Complete Prompts

Figures A14–A17 expand the four cases in Figure 5. Each case pairs the archived verification answer with the same MLLM's regenerated answer after verifier feedback, alongside the Direct-*G* candidate and ASV decision.

The MLLM heatmaps show attention to image tokens that predict the final yes/no answer, obtained by teacher-forced replay of each saved response under its exact user prompt. We average the final four decoder layers and all heads; InternVL visualizations use its full-image thumbnail tile. OmniVerifier evidence is displayed separately; the maps characterize recorded attention patterns (Jain and Wallace, 2019). Each map is divided by its image-patch mean, with a common 95th-percentile display limit across the three maps in each case; blue indicates lower and red higher relative attention. The full user prompts and complete generated answers accompany every case.

Direct-*G* case inputs. The spatial panels use the following complete user templates, with `{expr}` replaced by the query printed in each figure and the corresponding image provided as input.

Table A36: **InternVL3.5-8B Prompt.**

User message

<image>
Task: Decide whether the referring expression strictly matches a visible target in the image, then localize it if it exists.
Referring expression: "{expr}"
If the target exists, return exactly one bounding box in absolute image pixels as [x1, y1, x2, y2].
If the target does not exist, return exactly: not found.
Do not output explanations.

Table A37: **UniVG-R1-7B Prompt.**

User message

Please find "{expr}" in the image. Output the thinking process in <think></think> and final answer in <answer></answer> tags. If a matching object exists, output bbox as [x1,y1,x2,y2]. If no matching object exists, output not found.

Original spatial responses. For the Association, Attribute, and Relation cases, InternVL returns `[404, 360, 868, 998]`, `[237, 864, 705, 1000]`, and `[[288, 271, 650, 968]]`, respectively. UniVG's complete Object response is:

UniVG-R1-7B: saved response

<think>First, I need to identify the object described as "the robot in front of the pizza." I will look for a figure that resembles a robot and is positioned near the pizza. In the image, there are three people and a pizza on the table. The person on the left appears to be interacting with the pizza, so I will focus on them. This person has long hair and is wearing a purple top. They seem to be the one described as "the robot in front of the pizza." Therefore, the bounding box coordinates for this object should encompass the area where this person is standing.</think>
<answer>(10,12),(557,858)</answer>

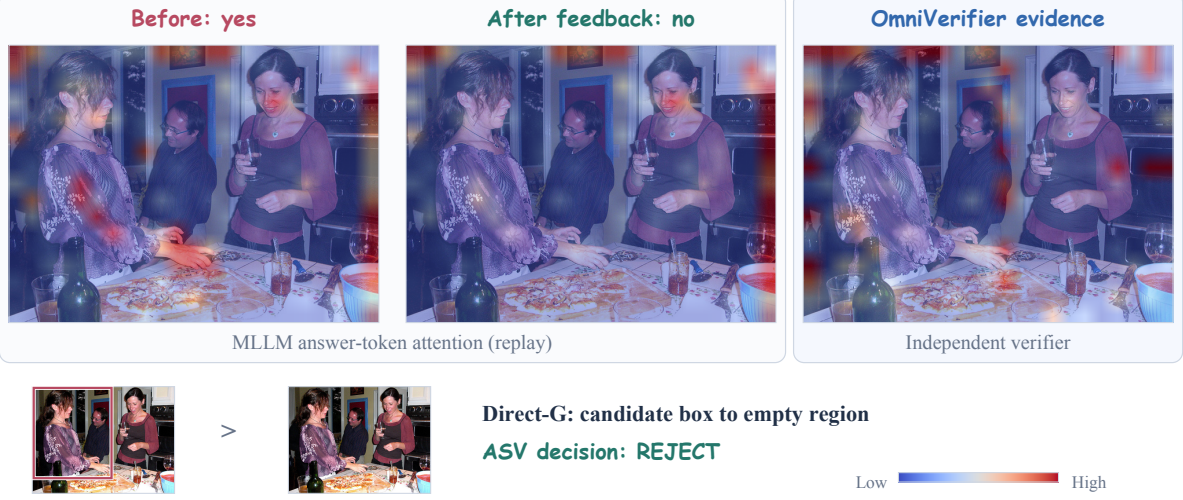
The returned coordinates are mapped from the models' $[0, 1000]$ output scale to image pixels for display; ASV returns an empty region in all four cases.

F LLM Usage

We used large language models (LLMs) to improve the clarity and grammatical correctness of the manuscript. After using these tools, the authors carefully reviewed and edited all generated content, and take full responsibility for the final version of the paper.

Object (EOH) — UniVG-R1-7B

Query: the robot in front of the pizza



Original user prompt

Please decide if "the robot in front of the pizza" strictly matches a visible target in the image. Output the thinking process in <think></think> and final answer in <answer></answer> tags. The answer should be exactly "yes" or "no".

Original complete output

<answer>yes</answer>

Feedback user prompt

Earlier answer: <answer>yes</answer>

An independent visual verification module inspected candidate region [6.4, 5.8, 356.5, 411.8] (pixel coordinates x1, y1, x2, y2) for the referring description "the robot in front of the pizza". Its actual candidate-level findings are: Omni support logit (true minus false; positive favors support) = -6.875; JEV calibrated support probability = 0.006; fixed ASV candidate decision = REJECT. These checks concern only this proposed region; another matching target may still exist elsewhere. Reinspect the whole image and answer the original question using visible evidence. You may retain or revise your earlier answer.

Original question: Please decide if "the robot in front of the pizza" strictly matches a visible target in the image. Output the thinking process in <think></think> and final answer in <answer></answer> tags. The answer should be exactly "yes" or "no".

Complete regenerated output

<think>First, I need to identify the reference mentioned in the source image described by the question. The question asks if "the robot in front of the pizza" strictly matches a visible target in the image. I should look for an object or area that resembles a robot and is positioned in front of a pizza. Next, I will search the target image for any area that meets this description. After examining the target image, I notice that there is no visible target that matches the description of a robot in front of a pizza. Therefore, the answer must be "no".</think>
<answer>no</answer>

Figure A14: **Object hallucination (EOH), UniVG-R1-7B.** Original and feedback prompts, complete answers, MLLM answer attention, independent verifier evidence, and the ASV grounding decision.

Association (EOH) — InternVL3.5-8B

Query: the colorful tent in front of the person

Before: yes | After feedback: no



MLLM answer-token attention (replay)

OmniVerifier evidence



Independent verifier



>



Direct-G: candidate box to empty region

ASV decision: REJECT

Low — High

Original user prompt

Task: Decide whether the referring expression strictly matches a visible target in the image.
Referring expression: "the colorful tent in front of the person"
The object identity, number, attributes, colors, and spatial relations must all match.
Answer with exactly one word: yes or no.

Original complete output

yes

Feedback user prompt

Earlier answer: yes

An independent visual verification module inspected candidate region [258.6, 153.7, 555.5, 426.1] (pixel coordinates x1, y1, x2, y2) for the referring description "the colorful tent in front of the person". Its actual candidate-level findings are: Omni support logit (true minus false; positive favors support) = -1.625; JEV calibrated support probability = 0.006; fixed ASV candidate decision = REJECT. These checks concern only this proposed region; another matching target may still exist elsewhere. Reinspect the whole image and answer the original question using visible evidence. You may retain or revise your earlier answer.

Original question: Task: Decide whether the referring expression strictly matches a visible target in the image.
Referring expression: "the colorful tent in front of the person"
The object identity, number, attributes, colors, and spatial relations must all match.
Answer with exactly one word: yes or no.

Complete regenerated output

No.

Figure A15: **Association hallucination (EOH), InternVL3.5-8B.** Original and feedback prompts, complete answers, MLLM answer attention, independent verifier evidence, and the ASV grounding decision.

Attribute (ROH) — InternVL3.5-8B

Query: the red dining table with pizza on it

Before: yes



After feedback: no



MLLM answer-token attention (replay)

OmniVerifier evidence



Independent verifier



>



Direct-G: candidate box to empty region

ASV decision: REJECT

Low — High

Original user prompt

Task: Decide whether the referring expression strictly matches a visible target in the image.
Referring expression: "the red dining table with pizza on it"
The object identity, number, attributes, colors, and spatial relations must all match.
Answer with exactly one word: yes or no.

Original complete output

yes

Feedback user prompt

Earlier answer: yes

An independent visual verification module inspected candidate region [151.7, 429.4, 451.2, 497.0] (pixel coordinates x1, y1, x2, y2) for the referring description "the red dining table with pizza on it". Its actual candidate-level findings are: Omni support logit (true minus false; positive favors support) = -4.375; JEV calibrated support probability = 0.003; fixed ASV candidate decision = REJECT. These checks concern only this proposed region; another matching target may still exist elsewhere. Reinspect the whole image and answer the original question using visible evidence. You may retain or revise your earlier answer.

Original question: Task: Decide whether the referring expression strictly matches a visible target in the image.
Referring expression: "the red dining table with pizza on it"
The object identity, number, attributes, colors, and spatial relations must all match.
Answer with exactly one word: yes or no.

Complete regenerated output

No.

Figure A16: **Attribute hallucination (ROH), InternVL3.5-8B.** Original and feedback prompts, complete answers, MLLM answer attention, independent verifier evidence, and the ASV grounding decision.

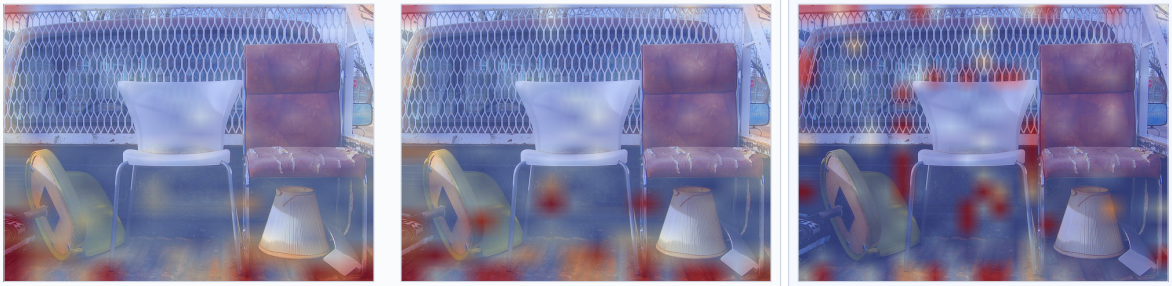
Relation (ROH) — InternVL3.5-8B

Query: the white chair under the brown chair

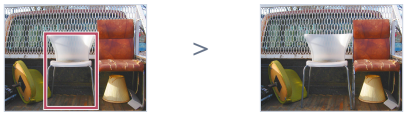
Before: yes

After feedback: no

OmniVerifier evidence



MLLM answer-token attention (replay) | Independent verifier



Direct-G: candidate box to empty region

ASV decision: REJECT

Low — High

Original user prompt

Task: Decide whether the referring expression strictly matches a visible target in the image.
Referring expression: "the white chair under the brown chair"
The object identity, number, attributes, colors, and spatial relations must all match.
Answer with exactly one word: yes or no.

Original complete output

yes

Feedback user prompt

Earlier answer: yes

An independent visual verification module inspected candidate region [184.3, 130.1, 416.0, 464.6] (pixel coordinates x1, y1, x2, y2) for the referring description "the white chair under the brown chair". Its actual candidate-level findings are: Omni support logit (true minus false; positive favors support) = 7.500; JEV calibrated support probability = 0.013; fixed ASV candidate decision = REJECT. These checks concern only this proposed region; another matching target may still exist elsewhere. Reinspect the whole image and answer the original question using visible evidence. You may retain or revise your earlier answer.

Original question: Task: Decide whether the referring expression strictly matches a visible target in the image.
Referring expression: "the white chair under the brown chair"
The object identity, number, attributes, colors, and spatial relations must all match.
Answer with exactly one word: yes or no.

Complete regenerated output

No.

Figure A17: **Relation hallucination (ROH), InternVL3.5-8B.** Original and feedback prompts, complete answers, MLLM answer attention, independent verifier evidence, and the ASV grounding decision.